Fabrication, characterisation and modelling of electronic devices based on amorphous metal-oxide and carbonaceous materials

A thesis submitted in fulfilment of the requirements for the degree of Doctor of Philosophy

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Declaration

I certify that except where due acknowledgement has been made, the work is that of the author alone; the work has not been submitted previously, in whole or in part, to qualify for any other academic award; the content of the thesis is the result of work which has been carried out since the official commencement date of the approved research program; any editorial work, paid or unpaid, carried out by a third party is acknowledged; and, ethics procedures and guidelines have been followed.

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Abstract

This thesis describes the deposition, characterisation and device applications of amorphous zinc tin oxide (a-ZTO) and carbon-based materials using energetic deposition techniques. The study aimed to explore the suitability of this growth method in producing high-quality carbon and a-ZTO films for device applications. Technology Computer Aided Design (TCAD) simulations were also conducted in tandem with experiments to identify critical parameters affecting device behaviour and to provide guidance for improved device performance.

Firstly, electrical carbon contacts to n-type 6H-SiC were energetically deposited from a filtered cathodic vacuum arc (FCVA) at room temperature and elevated temperature with low and high biases. Lower energy (< 100 eV) in the carbon flux resulted in resistive amorphous carbon contacts. As the deposition energy and sp^2 bonding (graphitic) fraction were increased, an oriented graphitic microstructure developed and rectifying electrical characteristics emerged. TCAD simulations revealed the effects of interfacial layers and contact work functions on the device performance and suggested that the rectification ratios of C/6H-SiC Schottky diodes could be increased by improving the lateral homogeneity of the junctions and/or by controlling the thickness of interfacial layers.

Secondly, thin films of amorphous zinc tin oxide (a-ZTO) were energetically deposited using high power impulse magnetron sputtering (HiPIMS). HiPIMS and DC magnetron sputtering modes were enabled to co-deposit an a-ZTO layer with Zn:Sn ratio that varied laterally across a 4-inch diameter sapphire substrate. Electrical, structural and optical properties

of the films were investigated as a function of composition. The as-deposited films were found to be amorphous, transparent and highly resistive with little variation in the Zn:Sn ratios. Annealing in the presence of hydrogen yielded improved film conductivity and measured carrier concentrations of $\sim 10^{17}$ cm^{-3}. Hall mobilities of up to 13 $cm^2/V.s$ were also measured in the n-type films. These findings suggest that HiPIMS can produce dense, high quality a-ZTO suitable for device applications.

As a transparent amorphous conducting oxide with high transparency and good electron mobility, a-ZTO has proven applications in interconnects and thin film transistors. In this thesis, the potential for this material in 'next-generation' signal processing devices is discussed. Specifically, the ability of the material to support resistive switching and memristive phenomena was investigated in lateral memristors on HiPIMS a-ZTO. The transport mechanisms and conductance of Ag/a-ZTO memristors were found to depend on prior activity and on the imposed current limit, mimicking biology synaptic plasticity. After microscopy, the switching mechanism was attributed to nanoscale filaments formed between the electrodes. These filaments were subject to Rayleigh instability and exhibited relaxation times determined by their effective radii.

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10) H. Pham, **H. N. Tran**, A. S. Holland, and J. G. Partridge, "Temperature-Dependent Electrical Characteristics and Extraction of Richardson Constant from Graphitic-C/n-Type 6H-SiC Schottky Diodes," *Journal of Electronic Materials*, pp. 1-6, 2019.

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Abbreviations

AFM	Atomic Force Microscopy
AOS	Amorphous Oxide Semiconductor
AOS	Amorphous Oxide Semiconductors
CNT	Carbon Nanotubes
C-Si	Carbon-Silicon
CVD	Chemical Vapor Deposition
DC	Direct Current
DCMS	Direct Current Magnetron Sputtering
DFT	Discrete Fourier Transform
DRAM	Dynamic Random Access Memory
EELS	Electron Energy Loss Spectroscopy
FCVA	Filtered Cathodic Vacuum Arc
FE	Fermi Energy
FEM	Finite-Element Method
FPGA	Field-Programmable Gate Arrays
HF	Hydrofluoric
HiPIMS	High Power Impulse Magnetron Sputtering
HRS	High- Resistance States
IGZO	Indium Gallium Zinc Oxide
ITO	Indium Tin Oxide
ITRS	International Technology Roadmap for Semiconductor
I-V	Current-Voltage
LRS	Low- Resistance States
MESFET	Metal-Semiconductor Field-Effect-Transistors
MIM	Metal-Insulator-Metal
M-O-M	Metal-Oxygen-Metal
MS	Metal-Semiconductor
NMOS	Negative Metal Oxide Semiconductor

P-F	Poole-Frenkel
PLD	Pulsed Laser Deposition
PMOS	Positive Metal Oxide Semiconductor
PVD	Physical Vapor Deposition
PyC	Pyrolytic Carbon
RF	Radio Frequency
RMS	Root Mean Square
RR	Rectification Ratio
RRAM	Resistive Random Access Memory
SBDs	Schottky-Barrier Diodes
SDE	Sentaurus Device Structure Editor
SDevice	Sentaurus Device
SEM	Scanning Electron Microscopy
SMesh	Sentaurus Mesh
Sprocess	Sentaurus Process
SRAM	Static Random Access Memory
SVisual	Sentaurus Visual
TCAD	Technology Computer Aided Design
TCOs	Transparent Conducting Oxides
TE	Thermionic Emission
TEM	Transmission Electron Microscopy
TFTs	Thin Film Transistors
UV	Ultra Violet
VBM	Valence Band Maximum
XPS	X-Ray Photoelectron Spectroscopy
XRD	X-Ray Diffraction
ZTO	Zinc Tin Oxide

Chapter 1
Introduction

1.1. Background

As electronics becomes more deeply embedded in everyday items, there is an increasing need to develop electronic materials that are flexible, durable and able to withstand variable environmental conditions [1-3]. In many applications, notably touch screens, transparency is also desirable. For instance, transparent displays are expected to increasingly appear in automobiles so that drivers can access traffic information from their immediate environment (smart windscreen) while driving a car [4]. Existing applications include high-resolution flat-panel displays and wearable electronics. These devices are required to be flexible, lightweight, shock resistant and importantly, affordable. The conventional materials for these applications are organic semiconductor and hydrogenated amorphous silicon (a-Si:H). However, the comparatively low mobilities of these materials (typically less than 5 $cm^2/V.s$ compared with > 200 $cm^2/V.s$ for crystalline Si) limit device performance. In addition, Si-based devices are not transparent unless the Si is sufficiently thin. This can induce damage in the Si and incurs additional cost.

Amorphous wide bandgap semiconducting materials fulfil some or all of the aforementioned requirements for transparent, flexible electronics and have received much attention in recent years [5,6]. Amorphous semiconductors generally have lower processing temperatures and offer greater uniformity of device characteristics when compared with

polycrystalline semiconductors. Gallium indium zinc oxide (GIZO) and indium tin oxide (ITO) are leading contenders among the amorphous oxide semiconductors (AOS) as they offer flexibility and reasonably high carrier mobility coupled with excellent transparency [7,8]. *Sharp* have commercialised GIZO thin film transistors, using them in their high resolution flat panel displays [9] and more recently GIZO has found its way into the *Ipad Air* [10]. However, commercial concerns over the scarcity and expected price volatility of indium have led investigators to assess other amorphous metal oxides. Zinc oxide (ZnO) has a distinguished history and has been exploited by scientists working in sensing, high frequency electronics and optoelectronics. Amorphous and nanocrystalline forms have been employed to produce transparent electronics (see for example [11]). When mixed with tin oxide (SnO_2) to produce amorphous zinc tin oxide (ZTO), this transparent conducting oxide becomes more resistant to chemical attack whilst maintaining properties suitable for electronics, including carrier mobility that exceeds that of amorphous Si.

Due to their necessity in any electronic device, electrical contacts to amorphous transparent conducting oxides have been the focus of a vast number of research reports and patents [12,13]. For the next generation of flexible transparent electronics, higher performance in the areas of flexibility, cost effectiveness and durability are sought. Two dimensional materials have been heralded as a game changer in many technological areas and have been employed in high performance flexible contacts [14,15]. To date however, compatibility with conventional CMOS (Complementary Metal Oxide Semiconductor) processing has not been demonstrated. This precludes large scale implementation and commercial viability. Importantly though, some of the excellent properties of the two-dimensional materials graphene and graphene oxide may be accessible in some thin film forms of carbon with high graphitic bonding fractions. In addition,

these can be patterned in CMOS compatible routes. This has been discussed in the literature but commercial applications for carbon thin films are in their infancy and have so far been largely restricted to protective layers.

Graphitic carbon offers several advantages over alternative contact materials. It is chemically inert and stable at extreme temperatures [16-18]. High quality graphitic-carbon contacts have been demonstrated on a variety of semiconducting materials including Si, 4H-SiC and GaAs [19]. It also has excellent thermal stability [20], making it suitable for power devices or high temperature environments. Finally, it exhibits sufficient conductivity and flexibility for many flexible electronic applications.

1.2. Research aims and objectives

Since Nomura et al. reported flexible GIZO based electronic devices in Nature in 2004 [21], the market for flexible electronic devices has grown enormously. Progress has been made in increasing the switching speed and reliability of transistors [22,23] but there remains room for improvement. In this thesis, two front-running materials are studied; amorphous ZTO and graphitic carbon. These electronic materials have both been synthesized using energetic deposition. ZTO offers an alternative to In-based flexible electronics and has demonstrated excellent carrier mobility and stability [24,25]. Energetically deposited ZTO has demonstrated one of the highest reported carrier mobilities suggesting its suitability for large- scale device arrays [26] but its application mainly involves thin film transistors. Similarly, graphitic carbon contacts have been fabricated using colloidal, low energy and high energy physical vapour deposition methods [27-29] but this project includes investigation of highly-oriented [30], graphitic contacts to semiconductors, including ZTO. The chemical, structural, optical and

electronic properties of these materials are studied using advanced microscopic and materials analysis methods. The materials are implemented in devices including Schottky barrier diodes and resistive switching devices (memristors). These experimental endeavours are supported by simulations to assist in identifying the most critical parameters in a particular device's behaviour and to suggest how improved performance may be achieved.

This study aims to answer the following research questions:

1) Can energetically deposited carbon be used for device contacts and does this method/material offer benefits over conventional methods/materials?
2) What are the critical processing and material parameters (eg. deposition energy and work function) affecting the properties of graphitic carbon-semiconductor junction?
3) Can energetic deposition produce a-ZTO of sufficient quality for device applications?
4) Can Technology Computer Aided Design (TCAD) be used to model devices based on a-ZTO and/or a-C?

The main outcomes of this research include:

1) High performance contacts to semiconductors produced by energetic deposition of carbon.
2) Electronic device layers produced from energetically deposited a-ZTO.
3) Improved understanding of the structure and performance limiting aspects of a-ZTO.
4) Controlled doping of a-ZTO by hydrogen was achieved supporting the assertion that hydrogen is a stable n-type dopant in ZnO and ZnO-based materials such as ZTO.
5) Accurate device models produced using TCAD finite element software. These revealed effects of lateral inhomogeneity of graphitic contacts to amorphous and crystalline semiconductors.

1.3. Thesis outline

This thesis describes the fabrication, characterisation and modelling of electronic devices based on amorphous zinc tin oxide and carbon-based materials. Chapters 4 - 6 of this thesis are adapted from peer-reviewed published scientific papers and the thesis is to be considered as *thesis based on publications*. The structure of the thesis is outlined as follows:

- Chapter 2 contains a literature review on the properties of the two materials studied in this research and some common methods used to grow these films.
- Chapter 3 describes techniques to deposit zinc tin oxide and carbon films, as well as to characterise their structural, optical and electrical properties.
- Chapter 4 studies the characteristics of junctions between graphitic carbon and n-type 6H-SiC. Finite element modeling of these junctions has been performed to identify critical parameters that most affect the device's behaviour and to provide guidance for improved performance. In this chapter, research questions 1 and 4 are addressed (outcomes 1 and 5).
- Chapter 5 details the deposition, characterisation and doping of a-ZTO using high power impulse magnetron sputtering (HiPIMS). This chapter addresses question 3 (outcomes 2-4).
- Chapter 6 investigates memristive switching behavior between Ag and a-ZTO. A planar structure was fabricated to reveal the filamentary nature of the device using electron microscopy and hence its capability to emulate biological synaptic behavior. This chapter provides research findings to answer question 3 (outcomes 2-4).
- Chapter 7 provides a summary on the work presented in this thesis and recommendations

for future work.

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Chapter 2

Literature Review

This chapter first presents an overview of transparent amorphous oxide semiconductors, followed by a more detailed discussion on the properties of zinc tin oxide. The properties of amorphous/disordered carbon are then discussed. Methods for growing thin films of these materials are described. Finally, basic metal-semiconductor junctions and resistive switching operation and structures are reviewed.

2.1. Amorphous oxide semiconductors

Transparent conducting oxides (TCOs) exhibit high bandgap (typically E_g > 3.1 eV), optical transparency (80 – 90 %), and conductivity [1]. TCOs are currently utilized in applications including thin-film solar cells and flat-panel displays. Indium gallium zinc oxide (IGZO) and indium tin oxide (ITO) are commercially chosen TCOs. Since indium is not as abundant as other metals, its cost is a concern. Approximately 1000 times more zinc (132 ppm) exists in the earth's crust than indium (0.1 ppm) [2,3]. Thus, efforts are being made to find TCOs that do not require In.

Amorphous semiconductors have lower processing temperatures and offer greater uniformity of device characteristics when compared with polycrystalline semiconductors. The traditional choice is a-Si:H but carrier mobility is typically lower by two orders of magnitude than that of single crystalline Si. By contrast amorphous oxide semiconductors offer comparable carrier mobilities to their corresponding crystalline phase.

Figure 2.1 illustrates the different carrier transport paths in a-Si:H and amorphous oxide semiconductor (AOS) containing post-transition-metal cations [10,11]. In a-Si:H, transport is controlled by hopping between localized tail-states and band conduction is not achieved. The transport path in this covalent semiconductor consists of sp^3 orbitals with strong directivity. Hence, fluctuations in the bond angles significantly alter the electronic levels, leading to reduced mobility. In contrast, degenerate band conduction and large mobility are possible in the AOS. Here, the bottom of the conduction band is mainly composed of spatially spread metal orbitals. The latter are isotropic in shape and exhibit direct overlap with neighbouring orbitals. The magnitude of this overlap is insensitive to the distorted metal-oxygen-metal (M-O-M) chemical bonds in AOS [12,13]. Furthermore, compared to binary oxide semiconductors, the structure of multicomponent oxide semiconductors tends to remain amorphous over a wide range of processing conditions. Zinc tin oxide (ZTO) is discussed below, as this material was the focus of much of the work in this thesis.

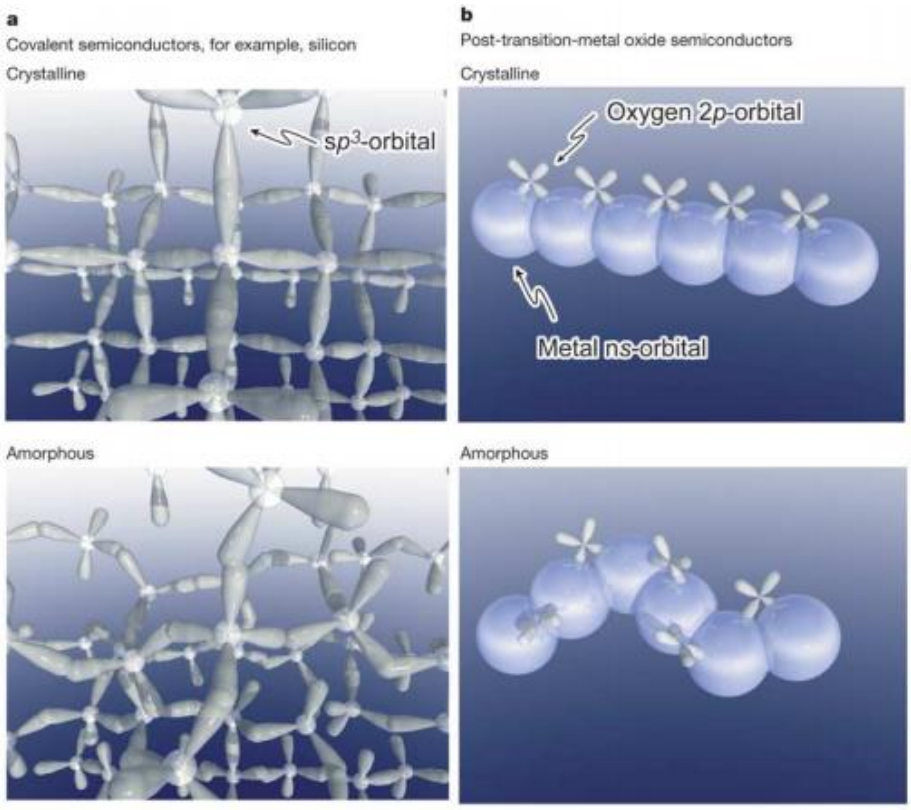


Figure 2.1: Schematic of orbitals determining carrier transport paths (that is, conduction band bottoms) in crystalline and amorphous semiconductors. (a) Covalent semiconductors with carrier transport paths composed of directional sp^3 orbitals and (b) amorphous oxide semiconductors composed of post-transition-metal cations with isotropically shaped overlapping orbitals (denoted by spheres here) [4].

2.2. Zinc tin oxide

ZTO is a wide bandgap (3.3 to 3.6 eV), n-type semiconductor [14]. There are two crystalline forms of ZTO, trigonal ilmenite ($ZnSnO_3$) [15] and cubic spinel (Zn_2SnO_4) [16]. In addition to transparency, aspects that make ZTO attractive are its low-cost, high chemical stability with respect to oxidation and etching, e.g. it withstands highly concentrated (> 35%) HCl solution [16,17], and excellent physical robustness [18].

Amorphous oxides composed of heavy-metal cations with $(n-1)d^{10}\ ns^0$ ($n >= 4$) electronic configurations exhibit high electron mobility [19,20]. Those vacant s-orbitals are spherical and exhibit a large spatial spread. Therefore, conduction can easily occur by a direct overlap of the s-orbitals in the neighbouring cations. In ZTO, Sn 4+ and Zn 2+ are such heavy-metal cations so amorphous ZTO (a-ZTO) exhibits comparable carrier mobility to its polycrystalline counterpart. This mobility is reportedly determined by the $5s^0$ orbital of Sn 4+ which is larger than the $4s^0$ orbital of Zn 2+ [21]. Provided suitable doping processes are available, applications for a-ZTO include transparent interconnects and active layers (channels) in transparent field effect transistors [19,21]. Transistors based on ZTO fabricated via sputtering show very good device characteristics, with mobility as high as 50 $cm^2/V.s$ and on/off ratios of 10^7 reported [19,22].

The properties of Zn-Sn-O films synthesized by sputtering and pulsed laser deposition (PLD) methods have been reported by Perkins et al [23]. Amorphous material resulted from both methods. In the sputter deposited films, the conductivity ranged from 10^{-2} to $10^{-1}\ \Omega^{-1}cm^{-1}$. These films were then annealed in N_2 at 625ºC and a polycrystalline phase was observed with X-ray diffraction peaks corresponding to Zn_2SnO_4. In these annealed films, the conductivity varied between 10^{-5} and $10^2\ \Omega^{-1}cm^{-1}$.

The structure of ZTO thin films deposited using PLD was investigated by Jayaraj et al [24]. The authors studied the film properties as a function of oxygen pressure and annealing temperature. The film thickness varied from 110 to 200 nm when the oxygen pressure gas was varied from 0 to 9 Pa. The deposited films were amorphous, but a crystalline phase was observed after annealing above 450º C. The optical bandgaps were 2.80 - 2.85 eV and were independent of oxygen pressure. The optimum Hall mobility and carrier concentration obtained from films deposited at low oxygen pressure (< 2 Pa) were ~ 10 $cm^2/V.s$ and ~ $10^{16}\ cm^{-3}$.

Niang et al. [25] investigated the influence of the stoichiometry of amorphous zinc tin oxide thin films employed as channel layers in thin-film transistors. The films were deposited using a remote-plasma reactive sputtering from zinc:tin metallic alloy targets with various tin compositions (10 %, 30 % and 50 %). The films were further optimised by varying the O_2 flow between 15 and 45 sccm. The authors reported the film thicknesses between 200 to 500 nm depending on deposition conditions with amorphous structure irrespective of composition. The optical bandgap increased from 3.5 to 3.8 eV with increasing Sn content. Average transmission of ~ 90% was observed in the visible range. Films sputtered with 25 sscm O_2 from 33% Sn target showed a Hall mobility of 5.7 $cm^2/V.s$, carrier concentration of 2.5×10^{17} cm^{-3}, and resistivity of 4 Ω.cm. Transistors produced from material deposited with these conditions achieved the highest performance, with threshold voltage, switching ratio, and field effect mobility of 8 V, 10^8 and 21 $cm^2/V.s$, respectively.

Aside from being employed in thin-film transistors [19,25], ZTO has applications in solar cells [26], as well as humidity and gas sensors [27-29].

2.3. Carbon-based materials

Carbon is one of the most abundant elements on earth and forms the basis of organic chemistry and life itself. Diamond and graphite are the most commonly observed forms of carbon. The properties of these materials are dramatically different because in graphite, the chemical bonding is trigonal (sp^2) and in diamond it is tetrahedral (sp^3) [30]. Diamond is an ultra-wide bandgap semiconductor and the hardest known material while semi-metallic graphite is conducting and soft enough to be used in pencils and dry lubricants. The ability to manipulate the electrical properties of carbon via its microstructure, sp^2:sp^3 ratio, dopant levels and/or by employing nano-dimensional forms has been exploited in active semiconducting device layers,

device contacts and/or interconnects [31-33]. Glassy- and pyrolitic- carbon are three dimensional forms with established applications and low production costs. Pyrolytic carbon (PyC) – a disordered nano-crystalline graphitic material, which can be formed through gas phase dehydrogenation (or pyrolysis) of hydrocarbons, exhibits good thermal and electrical conductivity as well as high durability [34]. As a result, PyC has been suggested for microelectronic applications including vertical interconnects, gate electrodes, and as a liner for trench capacitors in dynamic random access memory devices (DRAMS) [35-37]. For electrochemical applications, bulk glassy carbon has been used as an electrode material in micro-batteries due to its excellent thermal and electrical stability, large potential window, and low background noise [38,39]. Glassy carbon films created from pyrolyzed photoresist exhibit electrical conductivity that can be tuned by controlling the pyrolysis temperature. These have been employed as transparent conductors [40-42]. Colloidal graphitic carbon and PyC have been used to form Schottky contacts to several semiconductors [43,44]. Graphitic carbon has a work function ranging from 4.4 eV to 5.0 eV [45] and can therefore produce Schottky barrier with significant heights on n-type Si, GaAs and 4H-SiC [43]. In addition, graphitic-carbon contacts to semiconducting materials have demonstrated excellent chemical/thermal stability, mechanical durability and flexibility [46,47]. These properties suggest the potential for graphitic carbon as a high performance contact material.

The discoveries of the low-dimensional forms of carbon, nanotubes and later, graphene were swiftly followed by numerous reports on electronic applications for these materials (see for example [48-50]). The use of graphene in Schottky-Barrier diodes (SBDs) was reported by Chen et al. [51] among others and ideality factors approaching unity have been extracted from measurements, indicative of high-quality laterally homogeneous Schottky barriers. However,

implementation of graphene in large scale integrated devices has not been possible because, like carbon nanotubes and other low-dimensional materials, it cannot be assembled/patterned with precise dimensions/locations to form the large scale integrated device arrays necessary in most electronic equipment.

Thin films of carbon can be patterned using lithographic methods and can be grown using chemical vapour deposition or physical vapour deposition. The microstructure of any carbon film is complex but the sp^3:sp^2 ratio can be controlled within certain limits by growth temperature and/or by controlling the energy of the depositing species (atoms or ions) [52,53]. Perhaps the most widespread application of carbon thin films is as a protective layer on computer hard-drives. This diamond-like carbon is highly insulating and extremely hard meaning that only a very thin layer is required for protection. Applications for graphite-like films are scarcer but reports of highly oriented graphite-like films (as shown in Figure 2.2) have shown that it is chemically and thermally stable with high electrical conductivity and unique mechanical properties [54,55].

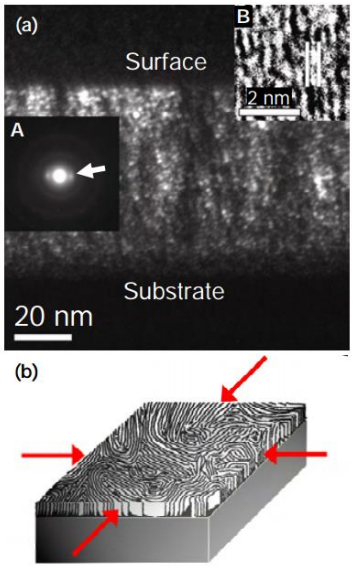
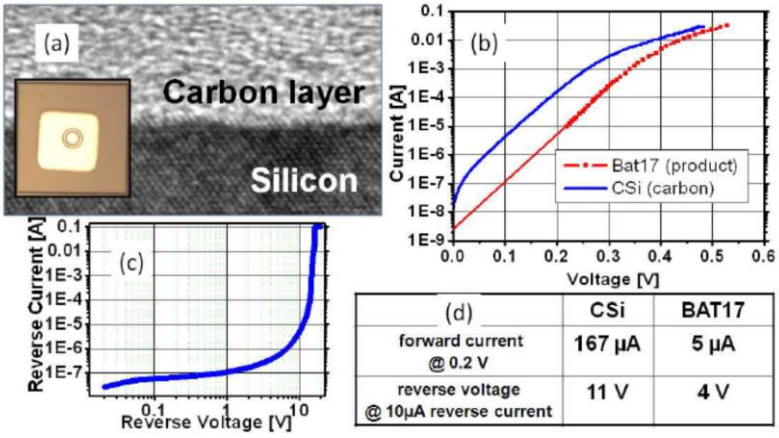


Figure 2.2: (a) Cross-sectional dark-field transmission electron microscopy (TEM) image of an oriented a-C film. This image was taken with an objective aperture centered on the graphitic {002} reflection (arrow in inset A). Inset B shows a high resolution image of the 0.335 nm graphite like planes (indicated by the vertical white lines) aligned perpendicular to the surface of the film. (b) Schematic representation of the arrangement of sp^2 sheets energetically preferred in a biaxial stress field indicated by arrows [54].

A study by Kreupl [56] showed that a Schottky diode formed with a carbon-silicon (C-Si) interface exhibited outstanding electrical characteristics and was able to withstand ultra-high current density. Schottky diodes are normally fabricated by depositing metals on n-type or p-type semiconductor materials to form metal-semiconductor contacts. In the event of high peak currents, diffusion of metal ions into the semiconductor occurs. As a result, degradation of the underlying junction, and device failure rapidly follow. To study the robustness of the carbon layer, a direct comparison on the electrical performance was performed between a C-Si diode and a commercial BAT17 diode from Infineon Ltd. All the device geometries, doping levels, and

guard ring implants were identical, only the metal Schottky contact was replaced by the carbon layer. Figure 2.3 shows that at a given voltage of 0.3 V, the forward current of the carbon diode is 15 times higher than of the BAT17. In the reverse bias, at low voltage, the C-Si diode allowed a higher leakage current through when compared with the BAT17 diode, due to a lower Schottky barrier height. At higher reverse voltages, it became lower than in the BAT17 diode, indicating that the breakdown voltage of the C-Si is higher; 11 V compared with 4 V for the commercial diode (BAT17) at a specified leakage current of 10 µA.



(d)	CSi	BAT17
forward current @ 0.2 V	167 µA	5 µA
reverse voltage @ 10µA reverse current	11 V	4 V

Figure 2.3: Comparison between C-Si and BAT17 commercial diodes (a) TEM cross-section of carbon deposited on silicon together with inset of a top-view of the circular diode area with Ti/Au metallization contact on top of a carbon layer. (b) The forward current characteristic. (c) The reverse characteristic of the C-Si diode and (d) a benchmark table [56].

As stated prior, C-Si interface diodes have also demonstrated excellent capability in withstanding high current levels [57], making carbon a promising candidate as a contact for power device applications. The high thermal stability of carbon was investigated by Stelzer and Kreupl [57] again by comparing with the commercial BAT15 TiSi diode. A high current density, around 3.5 MA/cm^2 with a pulse width of 100 ns, was applied to both diodes. After only 3

pulses, the TiSi diode showed a strong degradation due to an increase in reverse current. However, the C-Si diode was still fully functional after 100 million pulses, and interestingly, its reverse current reduced after these pulses, as shown in Figure 2.4.

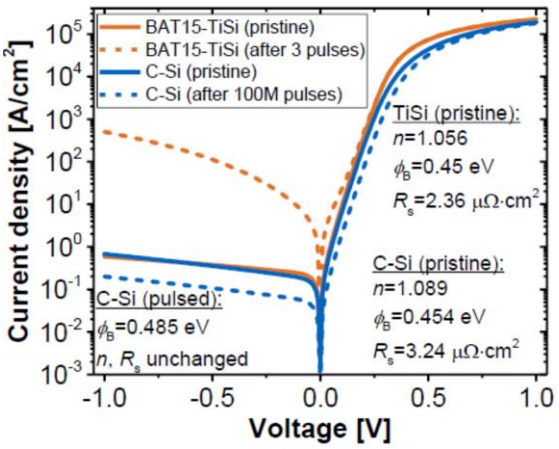


Figure 2.4: Comparison of the DC characteristics of the BAT15 (TiSi) and the C-Si diode with a carbon thickness of 58 nm after high current density pulses were applied [57].

2.4. Thin film ZTO and carbon deposition methods

To date, much of the work performed on flexible, amorphous materials has involved thin films. Here, the most widely used methods are briefly reviewed.

2.4.1. Chemical Vapor Deposition

Chemical Vapor Deposition (CVD) is widely used for high quality thin-films. In general, CVD involves flowing a precursor gas/gases into a chamber containing one or more heated objects to be coated. Chemical reactions occur on and near the hot surfaces, resulting in the deposition of a thin film. Operating temperatures range from 200-1600° C [58]. CVD has a number of advantages as a method for depositing thin films. In addition to the wide variety of materials that can be deposited, CVD films can be of very high purity because impurities can be

removed from gaseous precursors by distillation. Diamond films and carbon nanotubes (CNT) have been synthesized using CVD. With care, these can be grown in the same DC discharge CVD system by varying the pressure and composition of hydrogen-methane mixture and the discharge current and voltage [59].

Several factors limit the appeal of CVD. Precursors are often highly toxic, explosive or corrosive while byproducts of CVD reactions can also be very hazardous (CO or HF). Deposition temperatures are typically too high to coat temperature sensitive substrates such as flexible plastics.

2.4.2. Conventional (DC and RF) Sputtering

Magnetron sputtering is a Physical Vapor Deposition (PVD) process in which a plasma is created and positively charged ions that form the plasma are accelerated by an electric field towards a negatively charged “target”. The positive ions are accelerated by potentials ranging from a few hundred to a few thousand volts and strike the negative electrode with sufficient force to dislodge and eject atoms from the target. The most common sputtering techniques employed to form carbon films are DC and RF sputtering from a graphite target using an Ar plasma [60].

Sputtering is preferred for industrial applications because of its versatility and scalability. Also, the deposition conditions can be controlled by the plasma power and gas pressure, but they are reasonably independent of the substrate geometry or condition. Conventional sputtering produces relatively low energy ions and neutral species. This means that it cannot produce the best film adhesion and taking carbon as an example, it cannot produce the hardest diamond-like films or the highest conductivity graphite-rich oriented films that have potential applications as flexible contacts.

2.4.3. Energetic deposition

Figure 2.5 shows the generalized structure zone diagram for energetic deposition published by Andre Anders in 2010 [61]. This schematic plot follows on from an earlier (non-energetic) version published by Thornton in 1974 [62]. The figure shows the different film microstructures that can be created by adjusting the deposition temperature and/or energy. Crucially, energetic deposition enables some microstructures to be accessed at significantly lower temperatures than are possible with conventional low-energy deposition methods e.g. DC/RF sputtering and CVD. The additional energy that is required is typically gained by depositing ions from a plasma and simultaneously applying a DC or RF bias to the substrate on which the film is grown. If insulating substrates are used, the bias can also be applied to a conducting grid mounted in the plasma stream so that energetic neutrals are deposited [63]. Energetic deposition has been used to deposit high quality Al doped ZnO films for transparent interconnects and for a wide variety of amorphous and nanocrystalline carbon thin films including conducting oriented graphitic carbon [54,64].

Examples of energetic deposition methods include cathodic vacuum arc deposition and high impulse power magnetron sputtering (HiPIMS). In the former, a high current arc ablates a cathode and forms a plasma which can be steered magnetically to remove macroparticles which are generated in the explosive ablation process. In HiPIMS, a conventional magnetron sputtering target receives high power short duration pulses to create a dense plasma. Approximately ten times greater ionisation occurs when compared with conventional DC/RF sputtering.

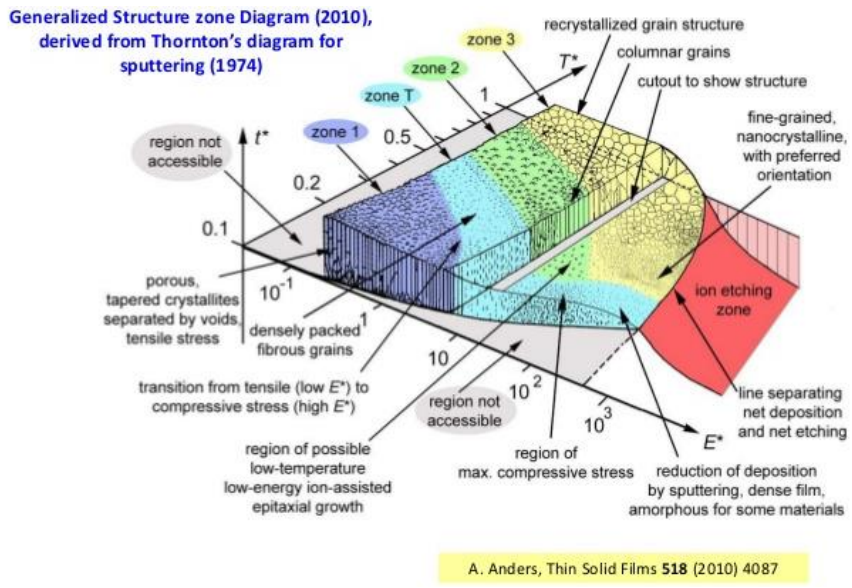


Figure 2.5: The generalized structure zone diagram for energetic deposition reported in ref. [61]

2.5. Metal-Semiconductor (MS) contacts

In many of the devices targeted for investigation in this project, metal-semiconductor junctions occur. These are briefly reviewed.

When a metal is in intimate contact with a semiconductor, the Fermi levels in the two materials must be aligned at thermal equilibrium and the vacuum level must be continuous across the interface. The metal work function is different from that of the semiconductor. From these conditions and for an ideal MS contacts where surface states are absent, the energy band diagram is shown in Figure 2.6

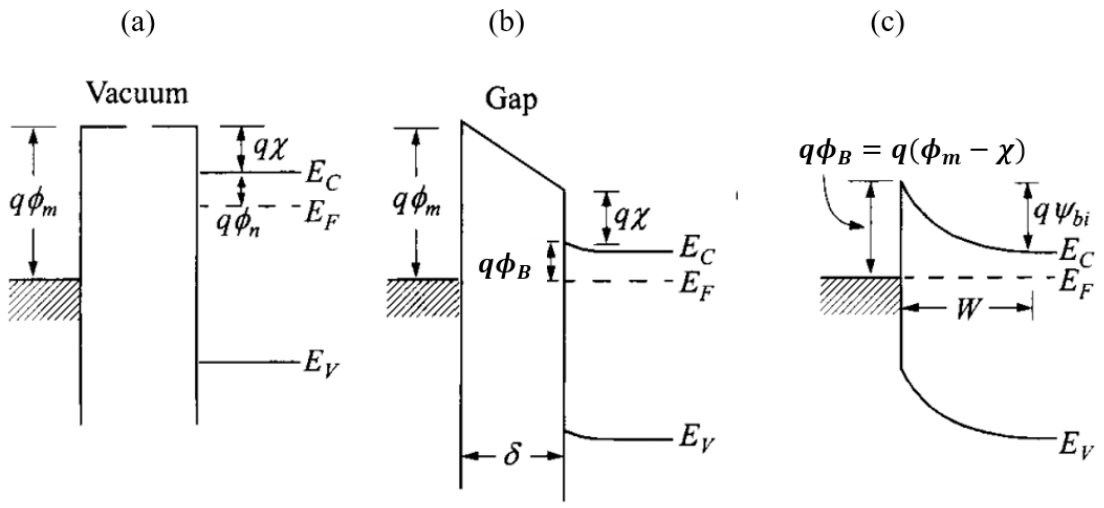


Figure 2.6: Energy-band diagram of MS contacts: (a) when separated; (b) connected into one system and (c) as the gap δ becomes zero [65]. The work function is the energy difference between the vacuum level and the Fermi level E_F. The quantity is denoted by $q\phi_m$ for the metal, and is equal to $q(\chi+\phi_n)$ in the semiconductor, where $q\chi$ is the electron affinity from the bottom of the conduction band E_C to the vacuum level, and $q\phi_n$ is the energy difference between E_C and the Fermi level.

There are four mechanisms that contribute to current transport across a MS contact: (i) thermionic emission (TE) over the potential barrier, (ii) quantum-mechanical tunneling of the carriers through the potential barrier, (iii) carrier recombination in the depletion region and (iv) carrier recombination in the neutral region of the semiconductor [66]. Ideally, the dominant contribution to the current transport comes from the TE of electrons over the barrier. For this ideal case, the potential barrier at the MS interface is simply the difference between the metal work function, ϕ_m and the semiconductor electron affinity, χ.

The ideality factor of a diode, *n* is a value describing how closely the diode follows the ideal behaviour. The equation for current transport *I* across the barrier is:

$$I = I_s \exp\left(\frac{q(V - IR_s)}{nkT}\right) \tag{2.1}$$

where I_s is the saturation current, V is the voltage across the diode, q is electron charge, k is Boltzmann's constant, T is temperature and n is the ideality factor.

The barrier height ϕ_B can be determined by extrapolating the semilog *I* versus *V* curve to *V=0*:

$$\phi_B = \frac{kT}{q} \ln \frac{(AA^*_{eff}T^2)}{I_s} \tag{2.2}$$

and the value of I_s and *n(kT/q)* are obtained from the slope and y-as intercept of the graph ln*(I)* versus *V* plot:

$$\ln(I) = \ln(I_s) + \frac{q}{nkT}V \tag{2.3}$$

$$n = \frac{q}{kT}\left(\frac{dV}{d(\ln(I))}\right) \tag{2.4}$$

where *A* is the active diode area and A^*_{eff} is the effective Richardson's constant.

In reality, other transport mechanisms often occur and the barrier height differs from that predicted by theory [67]. Causes for this are still debated [67] but the presence of surface states, interfacial layers and/or diffusion across the interface are all known contributors.

What is certain is that simple models can provide guidance in characterizing and designing Schottky junctions and are valuable in understanding the dominant transport mechanisms.

2.6. Memristor: the missing circuit element

A memristor is a two terminal device whose resistance depends on the magnitude, direction and duration of the applied voltage. When there is no applied bias, the memristor may remember its most recent resistance until voltage is reapplied. Hence, it can combine the behaviour of a memory and a resistor, hence its name: memristor. A typical current-voltage characteristic of a memristor is shown in Figure 2.7.

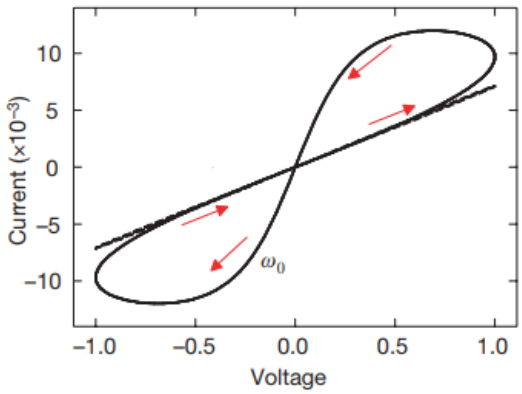


Figure 2.7: Current-Voltage (I-V) characteristic of a memristor [68]

The concept of memristor was first proposed by L. Chua in 1971 [68] as a fourth fundamental passive circuit element. He theoretically showed the relationship between magnetic flux and electric charge (see Figure 2.8). A memristor is characterised by its memristance function M, which is defined by the charge-dependent rate of change of flux with charge, as follows:

$$d\varphi = M \cdot dq \tag{2.5}$$

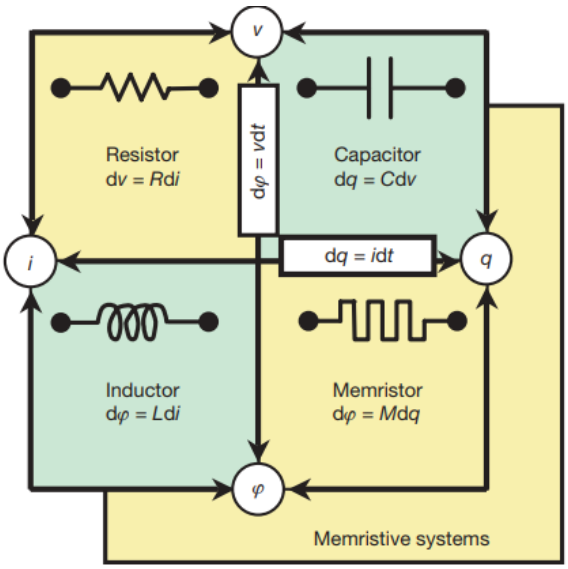


Figure 2.8: Memristor relationship between magnetic flux and charge, the missing connection among the four variables v, I, q and φ [69]

In 2008, thirty-seven years after Chua's theory was published [68], HP labs claimed to have "found" a memristor device. Their device consisted of a TiO_2 thin film sandwiched between platinum electrodes [70]. Since this work, it has become clear that memristance has been observed for more than 200 years [71]. The 5nm thick oxide in the HP device consisted of one lower layer of insulating TiO_2 and one upper layer of oxygen poor TiO_{2-x} (Figure 2.9). When a positive voltage was applied to the top electrode of the device, positively charged oxygen vacancies migrated towards and into the pure TiO_2 layer, thus decreasing its resistance. Applying a negative voltage produced the opposite effect with the oxygen vacancies attracted back to the top electrode, and the resistance of the TiO_2 layer returning to a value close to its original.

The HP team also presented a mathematical model for their memristor. This model was based on two series resistors with values R_{on} and R_{off} representing the doped and the undoped region resistances, D was the width of the device and w was the width of the doped region, as

shown in Figure 2.9. The doped region, *w*, represents the state variable which changes depending on the charge. The effective I-V behavior of the structure can be represented by the following equation:

$$v(t) = R_{on} * \frac{w(t)}{D} + R_{off} * \left(1 - \frac{w(t)}{D}\right) \tag{2.1}$$

And the memristance function, for $R_{on} \ll R_{off}$, is expressed by:

$$M(q(t)) = R_{off}\left(1 - \frac{\mu_v R_{on}}{D^2} q(t)\right) \tag{2.2}$$

where μ_v is the average mobility of dopants in the thin film.

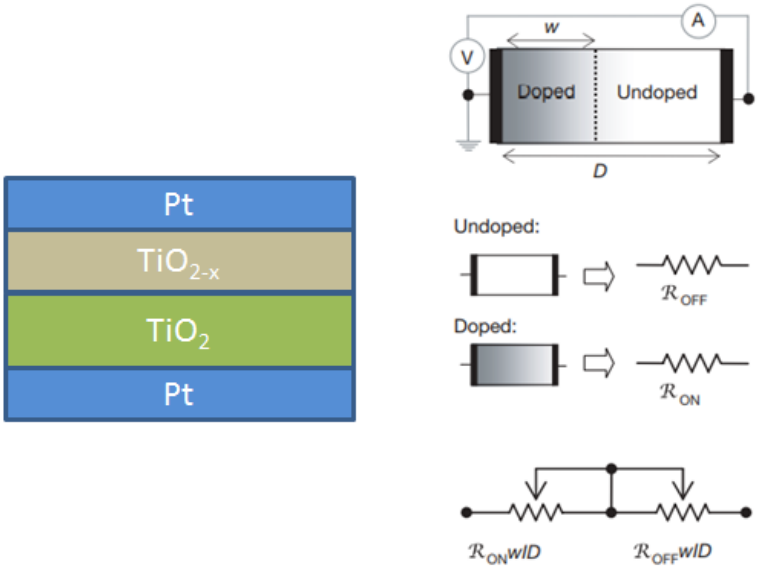


Figure 2.9: The HP-labs memristor and its simulation model reported in ref. [70]

2.7. Memristor-based applications

2.7.1. Analog circuit applications

Potential analog applications of memristor consist of programmable analog circuits, filters and chaotic circuits. Conventional programmable analog circuits require larger chip areas compared to digital circuits. By using memristors as digital potentiometers that operate under threshold conditions, it provides additional functions but consumes less chip area [72,73].

Memristors can improve circuit performance and provide methods to replace complex active components. The introduction of memristors to amplifier circuits enables variable gain with an increased linear range [74,75]. Memristor-capacitor (or inductor) combinations can provide non-exponential charge decay with multiple time scales. A memristor-capacitor-inductor circuit can be tuned from overdamped response to underdamped response by simply changing the polarity of the capacitor [76]. With a memristor-based bandpass filter, when input signals with strong components pass the filter at resonant frequency, the memristor experiences the highest input voltage. Hence its resistance reduces and provides a sharpening quality factor for the bandpass filter. The chip area and complexity of memristor-based filters are also substantially reduced as compared to the conventional resistor-capacitor filters [77].

Memristors also are potential for application in chaotic circuits for secure communications, random number generators, encryption systems and medical devices [78-84].

2.7.2. Digital logic applications

Memristors have been widely adopted in digital logic applications, such as combinations of memory and logic [85], logic gates [86,87] and field-programmable gate arrays (FPGA) [88].

A nonvolatile synchronous flip flop with an error rate of 0.1% was presented in ref [89]. A discrete FPGA architecture with memristor based interconnection showed 5.5 times improvement in area reduction, 2.3 times faster processing speed and power consumption reduced by 1.6 times [90].

2.7.3. Neuromorphic circuits

A typical neuromorphic computation system consists of neuron circuits and synapses. The synapses are the connections between neuron circuits. Each neuron may have more than 1000 synapse connections with other neurons (as shown in Figure 2.10). Scientists have simulated brains of small animals (cat, rat and spider) [91-93], where powerful computation resources were required. For example, to perform a simulation of a sub-set of processes in a cat brain, IBM's supercomputer BlueGene was used. It consists of clusters of 147456 microprocessors and 144 TB of memories, which consumed a power of 1.4 MW [94]. A cat's brain runs on less than 60 W. Therefore, the major challenge for a hardware implementation of a neuromorphic circuit is to develop electronic synapses that have high density and low energy consumption. The potential of resistive switching devices to function as synapses has been demonstrated by numerous groups [95-99]. Neurons essentially have two terminals, the pre-synaptic neuron and post-synaptic neuron, that are similar to the structure of memristors, as can be seen from Figure 2.10. In addition, memristors have also emulated synaptic plasticity in exhibiting gradual changes of their conductance. In biology, synaptic plasticity plays a vital role in facilitating learning and remembering in the brain.

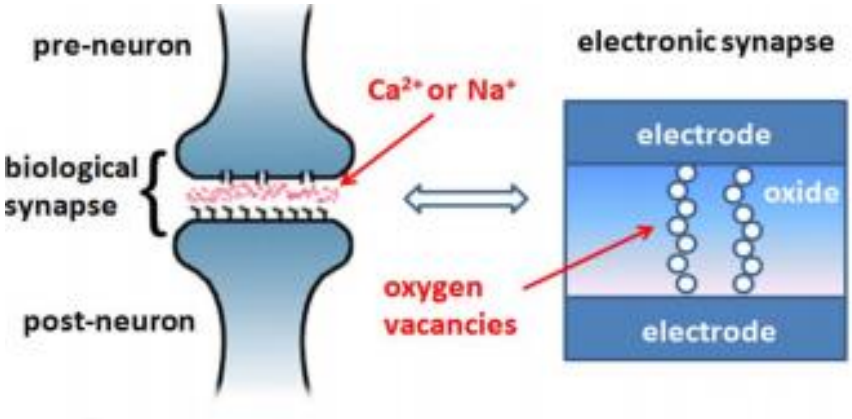


Figure 2.10: Schematic illustration showing biological synapse and a resistive switching device utilized as an artificial biological synapse between neurons [99]

2.7.4. Memories

Another potential application for memristors is Resistive Random Access Memory (RRAM). Traditional transistor-based memories such as Static Random Access memory (SRAM), Dynamic Random Access Memory (DRAM), and Flash have encountered scaling challenges. In 2013, a 32-Gb RRAM memory device using 24 nm CMOS technology was reported at the International Solid-State Circuit Conference (ISSCC) [100]. In 2020, it is expected that RRAM technology can expand the capacity up to 1 Tb and provide read/write access times of less than 100 ns with endurance in the order of 10^{12} cycles [101]. Compared to Flash memory which has slow write/erase process and low endurance cycles, RRAM offers similar memory capacity, but affords better in device performance such as speed (read time ~ 1 ns, write/erase time < 1 ns), power consumption, data retention (> 10 yrs) and endurance (> 10^{12} cycles) [102]. In addition, the crossbar-based architecture of RRAM is highly scalable and shows potential for future ultra-dense memory [103]. For example, RRAMs with feature sizes of 10 nm and 3 nm yield possible storage densities of 250 Gb/cm^2 and 2.5 Tb/cm^2, respectively [104]. The

32Gb RRAM chip already implemented by Sandisk consumed 1.3 cm^2 in area, with a memory density of 24.5 Gb/cm^2 [100].

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Chapter 3
Methodology

This chapter describes the techniques used to deposit amorphous ZTO and carbon thin films. Following this, experimental methods used to characterize these films, including optical, X-ray and microscopic techniques, are discussed. Next, thin-film and device measurements are presented. Finally, modelling of the devices is also discussed.

3.1. Energetic deposition of ZTO and carbon thin film

3.1.1. Filtered cathodic vacuum arc (FCVA)

In FCVA, a very high current (~ 20 - 200 A) and a low voltage (~ 20 V) are employed to create an arc discharge between an anode and a cathode. An almost fully ionised plasma is thus created from a cathode (target) material [1]. The plasma expands rapidly into an ambient and towards the substrate where the flux condenses to form a thin film. The plasma ions have kinetic energies ranging from 20 to 200 eV. As a result, this method provides films which are more dense and more strongly adhered than films deposited non-energetically. The ions penetrate several hundred Angstrom down through the substrate surface. One disadvantage of cathodic arc deposition is that “macroparticles” are produced and these degrade film quality [2,3]. The removal of macroparticles is generally accomplished using magnetic filters (as shown in Figure 3.1), that electromagnetically steer the deposition plasma away from the straight-line trajectories

followed by the heavier macroparticles. In the system used to deposit a-ZTO and a-C for this work, a double-bend magnetic filter was installed.

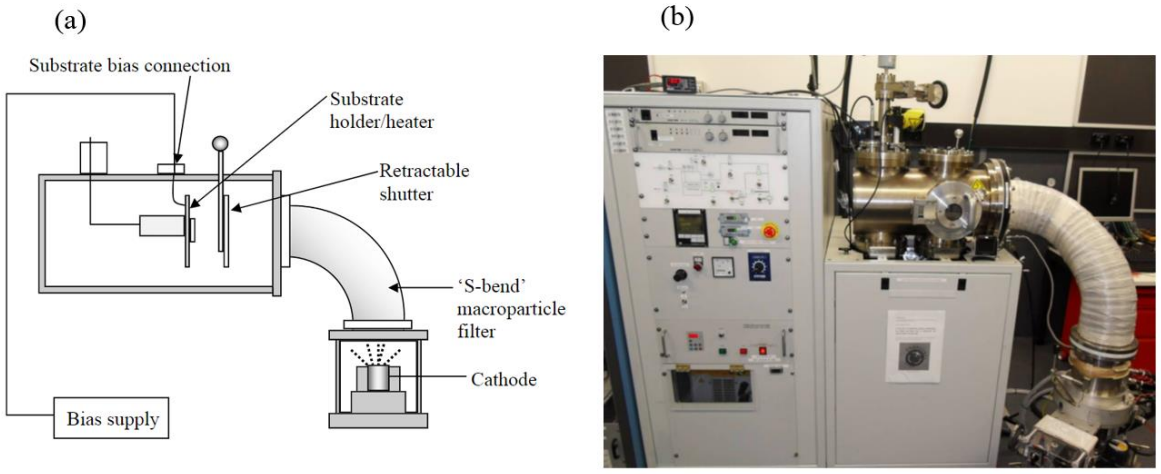


Figure 3.1: (a) Schematic illustration of a double-bend cathodic vacuum arc deposition system [4] and (b) photograph of the FCVA deposition system at RMIT University

In this work, FCVA deposition was used to grow a-ZTO and carbon device layers. The system was operated at a base pressure of $< 5\times10^{-6}$ Torr with a cathode current of 60 A. The carbon films were deposited using different combinations of bias voltage (~ 75 to 1000 V) and temperature (21 to 100° C), resulting in the formation of films ranging from insulating t-aC (tetrahedral amorphous carbon) to conducting graphite (details can be found in Chapter 4).

3.1.2. High power impulse magnetron sputtering (HiPIMS)

HiPIMS is an ionised physical vapor deposition technique which was first introduced in 1999 by Kouznetsov et al [5]. It uses similar apparatus to direct current (DC) or radio frequency (RF) magnetron sputtering but pulsed biasing at high power and low duty cycle is applied to the

sputter head. Due to its convenience and stability, it has attracted significant interest, especially in the field of industrial hard coatings [6].

In HiPIMS, a high power is applied to the magnetron (target) in the range of a few kW/cm^2 in very short pulses (~10 – 500 μs) [7,8], with a pulse frequency from 10 Hz to several kHz [9,10]. The voltage during the pulse in most HiPIMS processes is usually around 500 – 1000 V and the peak current density reaches a few A/cm^2. As a result, power densities at the target surface created in HiPIMS are more than two orders of magnitude higher than those typical in DC sputtering. This significantly increases the ionisation of the sputtered material. As in cathodic arc deposition, the high degree of ionisation leads to the growth of smooth and dense films and also enables more control over the phase composition, microstructure, mechanical and optical properties of the compound films [11-16]. It has also been reported to be beneficial in terms of improving conformality, enabling deposition of uniform films on different substrate shapes [17]. It can also reduce deposition temperatures since the duty cycle is small (< 10%) [18-20].

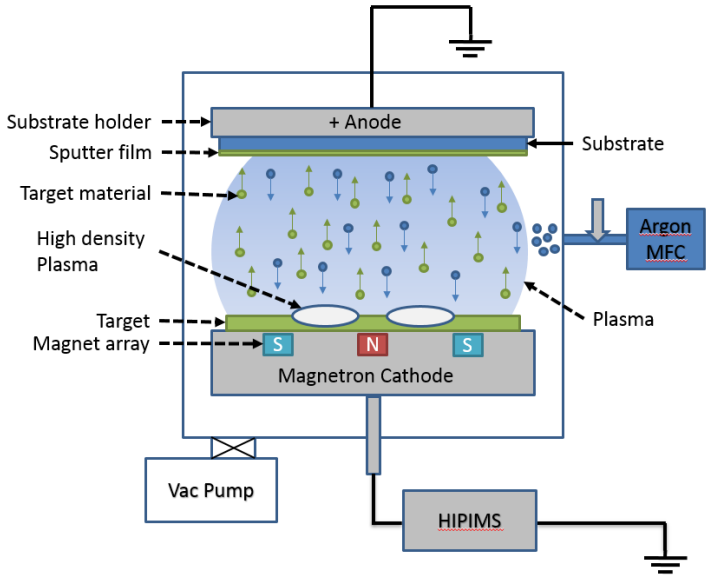


Figure 3.2: Schematic of the HiPIMS deposition system

In this project, the HiPIMS technique was used to grow a-ZTO films. The a-ZTO film was codeposisted using an AJA sputter deposition system and RUP-7 pulsed power supply. This system was equipped with four sputter heads, DC, RF and HiPIMS power supplies. The Zn target was operated in HiPIMS mode while the Sn target was operated in DCMS (Direct Current Magnetron Sputtering) mode, in the presence of Ar and reactive O_2 gases. More details on the deposition conditions can be found in Chapter 5.



Figure 3.3: Images of the HiPIMS system used to grow ZTO films for this work at the University of Sydney

3.2. Thin-film characterisation techniques

In this project, advanced material characterization was undertaken to understand the influence of synthesis parameters on structural, optical and electronic properties of the deposited films. These methods are now described.

3.2.1. X-ray photoelectron spectroscopy (XPS)

X-Ray photoelectron spectroscopy (XPS) is a technique for analysing the surface chemistry of a material. XPS can measure the elemental composition, chemical state and electronic state of the elements within a material. XPS spectra are obtained by irradiating a solid surface with a beam of X-rays while simultaneously measuring the kinetic energy of the

electrons that are emitted from the surface (to a depth of 10 nm) of the material being analysed (see Figure 3.4)

The kinetic energy (*K.E*) of the ejected electrons depends upon the photon energy (*hf*) and the binding energy (*B.E*), i.e. the energy required to remove the electron from its bound state. The binding energy depends upon a number of factors, including the following:

- The element from which the electron is emitted.
- The orbital from which the electron is ejected.
- The chemical environment of the atom from which the electron was emitted.

Many technologically important semiconductors have surfaces that differ greatly form their bulk. XPS is invaluable in studying these materials and in particular, can reveal important features within their bandgaps and around their valence band maxima. For example, surface band bending due to adsorbed OH has been observed in ZnO [22] and ZTO [23] using valence-band XPS and this has implications for the characteristics of junction devices formed on these materials, e.g. Schottky barrier diodes and ohmic contacts.

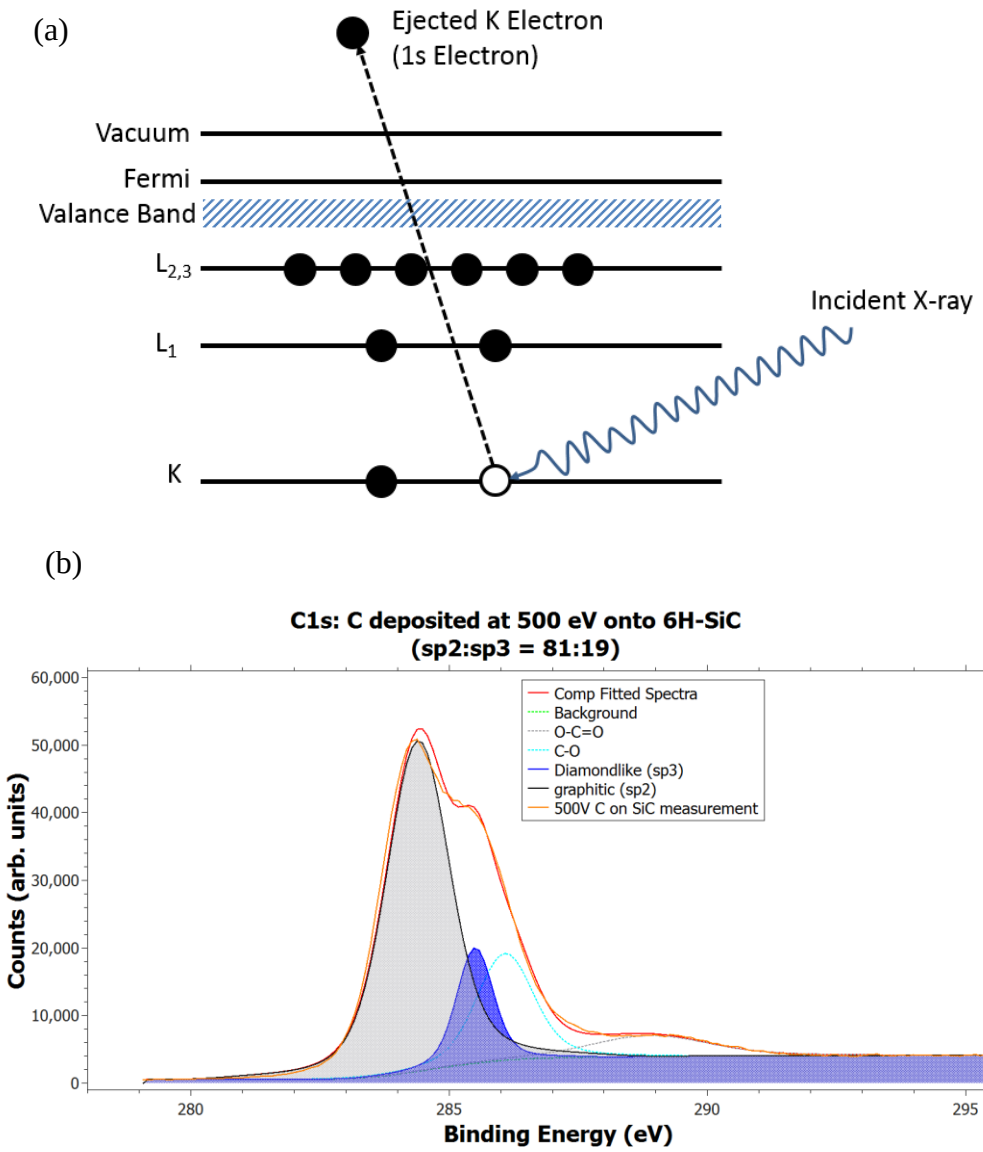


Figure 3.4: (a) The photoemission process involved for XPS surface analysis. The disc represents electrons and the bars represent energy levels within the material being analyzed. The equation governing the process is $K.E = hf - B.E$ [21]. (b) sp^2 fraction of the C films determined from X-ray photoelectron C 1s spectra

XPS analyses in this work were performed using a Thermo Scientific K-Alpha Spectrometer with an Al K-α source of 1486.6 eV energy. The spot size was ~ 400 µm for the

measurements. For semi-insulating substrates, a flood gun was enabled to minimize the charging effects during analysis. Depth profiles were also performed with the Ar ion beam of 200 eV, to lightly remove surface contaminants, such as OH attachment on the ZTO samples.

3.2.2. Scanning electron microscopy (SEM)

In SEM, a beam of electrons is produced by heating a metallic filament, such as W or LaB_6, and these electrons are attracted towards the anode [24]. The electron beam then follows a vertical path through the evacuated column of the microscope through a set of electromagnetic lenses which help to focus and direct the beam to the sample.

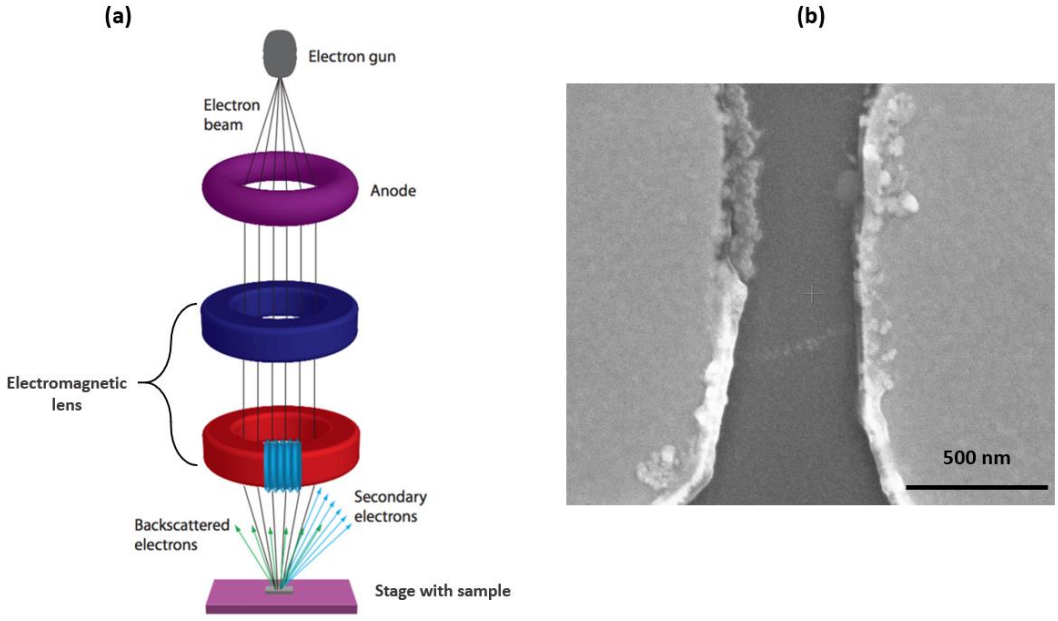


Figure 3.5: (a) Schematic of a SEM column showing the electron gun and lenses [25]. (b) SEM images showing Ag nanoparticle filament in Ag/ZTO/Ag memristor taken using a FEI SCIOS at 5.00 kV.

Once the electron beam interacts with the sample, backscattered and secondary electrons are ejected from the sample. With a standard detector, such as the Everhart-Thornely or Inlens detector, both backscattered and secondary electron signals can be collected [25] to provide information about the surface and the subsurface (to a depth of a few angstroms) [26]. In this thesis, SEM images shown were obtained using a FEI SCIOS SEM operation at 5.00 kV.

3.2.3. Transmission electron microscopy (TEM)

In a transmission electron microscope, a high energy beam of electrons is shone through a very thin sample (< 100 nm), and the transmitted electrons are used to observe features such as crystal structure and specimen orientations. The beam of electrons sourced from an electron gun is focused into a small, thin, and coherent beam by the use of condenser lens. This beam is restricted by the condenser aperture, which excludes high angle electrons. The part of the beam transmitted through the sample is focused by the objective lens into an image. Chemical analysis is also possible; the energies of the transmitted electrons can be measured using an electromagnetic prism and composition/bonding information gathered.

When the electron beam passes through the thin sample, it is diffracted by the atomic planes in the sample when the Bragg condition is satisfied [27]. These waves interact constructively and are brought to focus at the back focal plane of the objective lens to form diffraction patterns. In a disordered sample, since the crystal planes are in many directions, the diffraction pattern consists of concentric rings.

In this project, a JEOL 2100F TEM operated at 200 kV was used to image cross-sectional carbon and ZTO.

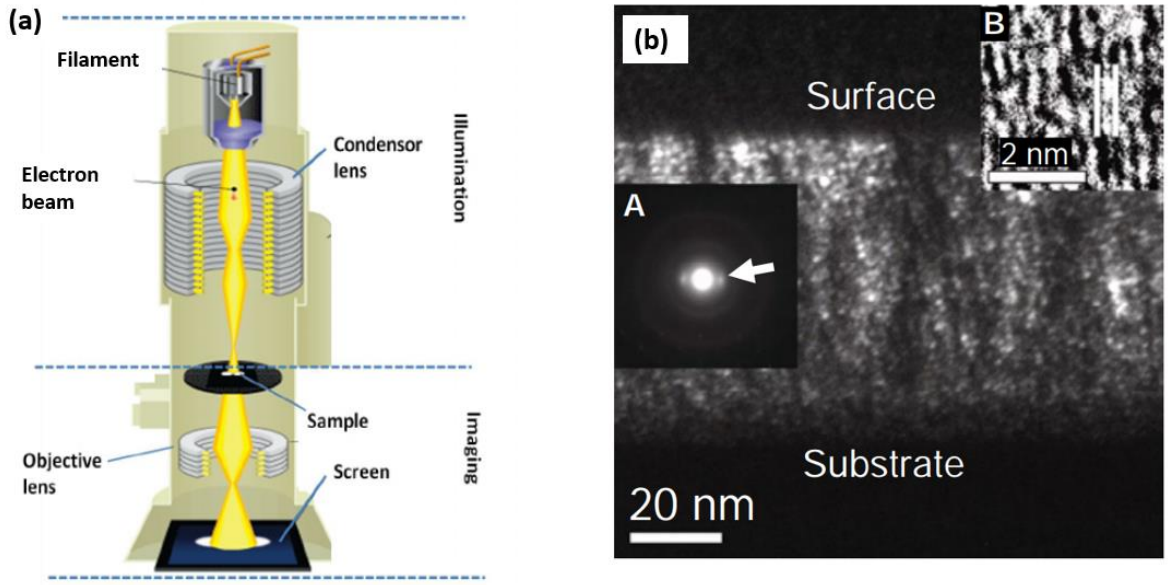


Figure 3.6: (a) Schematic diagram of a TEM [28]. (b) Cross-sectional dark-field TEM image of an oriented a-C film. This image was taken with an objective aperture centered on the graphitic {002} reflection (arrow in inset A). Inset B shows a high resolution image of the 0.335 nm graphitelike planes (indicated by the vertical white lines) aligned perpendicular to the surface of the film [29].

3.2.4. Atomic force microscopy (AFM)

AFM is a technique enabling the surface morphology of samples to be studied [30]. Here, the topography is determined from the deflection of a soft cantilever with a sharp tip mounted at its free end. As a tip scans across the sample surface, it is repelled or attracted by the intermolecular interactions (forces) between the atoms within the tip and the sample. As a result, the cantilever is deflected. The magnitude of the deflection is then obtained by a detection system consisting of a laser beam (at the end of the cantilever) and a photodiode (a so-called optical lever arrangement). A feedback mechanism from the detector is used to maintain either a

constant force or a constant deflection between the tip and the sample surface. The signal applied as feedback is converted into a Z-height and/or phase map.

In this research, a Veeco D3100 AFM was used to determine sample surface roughness and measure the film thickness from step-height measurements.

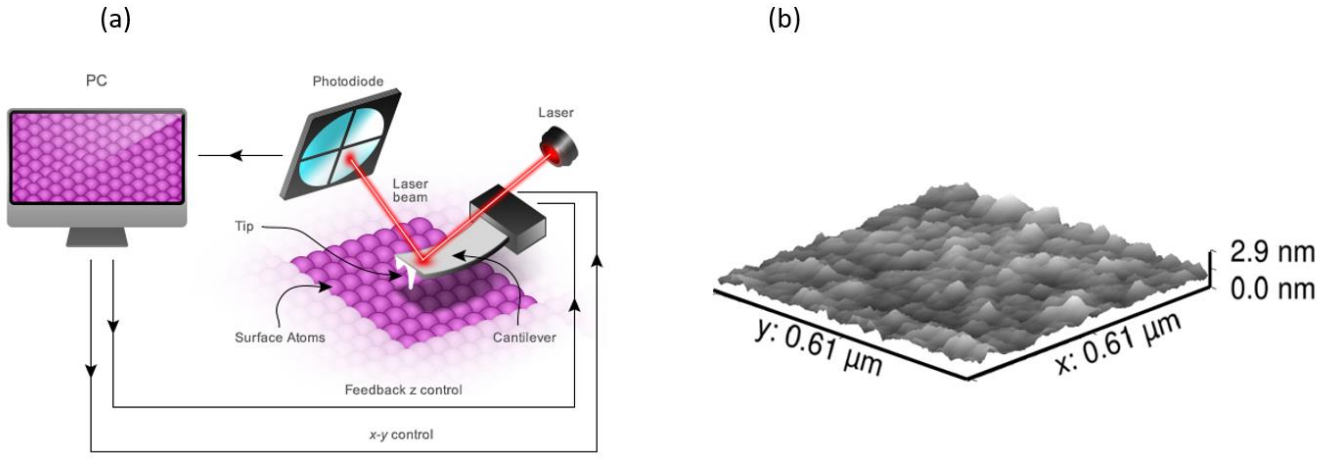


Figure 3.7: (a) A schematic illustration of an atomic force microscope connected to a computer [30]. (b) AFM image showing the surface morphology of the as-deposited a-ZTO film.

3.2.5. X-ray diffraction (XRD)

The goal of the XRD method is to identify crystal structures within elemental and compound samples. As in electron diffraction (Section 3.2.3), when the X-rays are Bragg peaks at angles θ of the diffracted beams from a crystal based on the constructive interference of scattered waves. This means that the diffracted rays are found only when the reflections from the parallel planes of atoms are in phase. It can be occurred if the difference in path length is equal to an integer times the wavelength λ. The condition for constructive reflection of the incident beam is verified by the Bragg law:

$$2d\ sin\ \theta\ =\ n\lambda \tag{3.1}$$

where *n* is the order of diffraction and *d* the interplanar spacing.

The measured intensity as a function of the scattering angle 2θ yields information of the spacing between planes of atoms in the crystal structure, as well as the sites of the atoms within the unit cell [31].

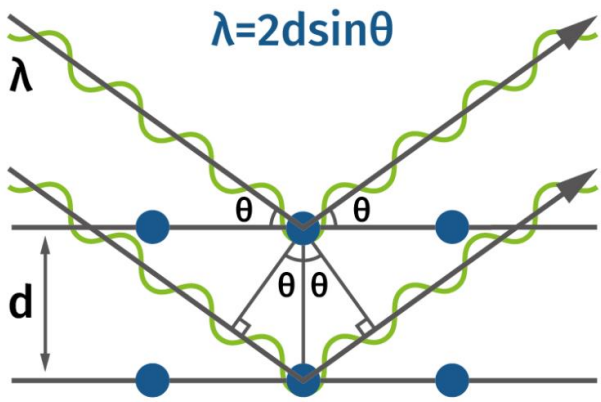


Figure 3.8: Schematic representation of Bragg's Law [32]

In this project, XRD studies were carried out on a Bruker D4 Endeavour diffractometer, using a monochromated Cu K-α source of wavelength λ = 1.54 Å and Bragg-Brentano configuration. X-ray intensity is recorded over a range of $2\theta = 20^o - 120^o$.

3.2.6. Optical transmission

When an incoming photon with energy $E_{ph} = hv$ is absorbed in a material, it can excite a valence electron to a higher energy state, i.e. conduction band. The information extracted from this absorption process can be used to analyse the bandgap of the material and the type of electronic transition.

The optical transmission across the appropriate wavelength spectrum, typically in the visible range from 250 nm to 850 nm is measured and from this data, the absorption coefficient, alpha, can be determined.

$$\alpha = \frac{1}{t}\ln\left(\frac{1}{T}\right) \tag{3.2}$$

where *t* is the thickness and *T* is the optical transmission of the film.

By plotting α^n *vs hv* (where *n* is 2 and ½ for direct and indirect transitions, respectively), the transition type and bandgap can be determined. The transition type is determined by the linearity of the fit near the absorption edge whilst the bandgap is determined by extrapolating α^n to the x-axis (*hv*).

For degenerately doped semiconductors, it is possible for the absorption edge to shift to higher energies, which is known as the Burstein-Moss shift [33,34]. Considering a direct bandgap semiconductor,

$$\alpha \sim \left(hv - E_g - \zeta_n\right)^2 \tag{3.3}$$

where ζ_n represents the depth the Fermi level is within the conduction band.

3.3. Electrical measurements

3.3.1. Resistivity and Hall measurement

The Hall effect, which was discovered in 1879, can be used to determine the resistivity, carrier type and concentration, and also mobility of a sample [35,36]. The Hall effect is based on the Lorentz force, as shown in Figure 3.9:

$$F = q(v^{\rightarrow} \times B^{\rightarrow}) \tag{3.4}$$

where q denotes the charge of the particle, v the velocity, and B the applied magnetic field.

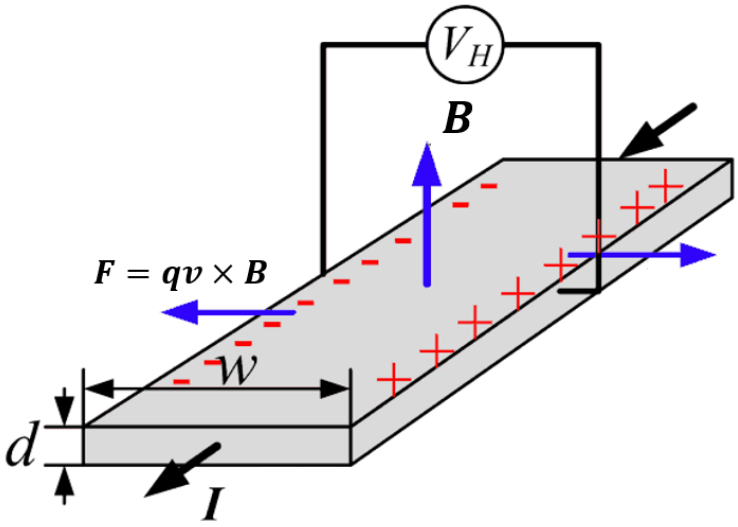


Figure 3.9: Principle of the Hall effect

The Lorentz force deflects carriers (with direction determined by the right-hand rule) towards one edge of the sample. A measurable potential, known as the Hall voltage, V_H, is then created across the sample:

$$V_H = \frac{BI}{qtn} \tag{3.5}$$

where I is the current, t is the sample thickness and n is the carrier concentration. The Hall coefficient, R_H, is defined as:

$$R_H = \frac{tV_H}{BI} \tag{3.6}$$

Hall effect measurements were performed on samples with Van der Pauw contact geometries (see Figure 3.10) using an Ecopia HMS 3000 system with a fixed magnetic field of 0.55T.

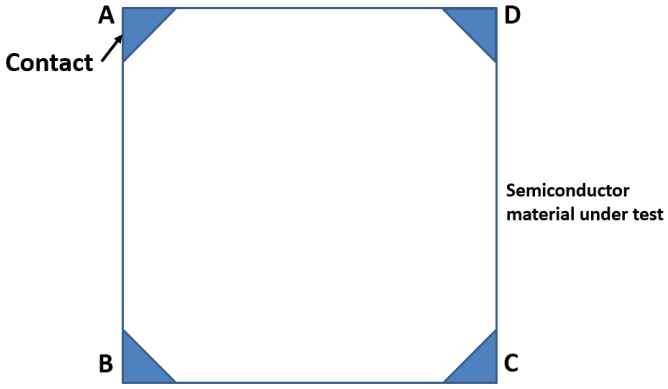


Figure 3.10: Van der Paww contact layout for sample in Hall measurement

Film resistivity is assessed using:

$$\rho = \frac{\pi t}{\ln(2)} \frac{R_{ab,cd} + R_{ac,bd}}{2} \tag{3.7}$$

$R_{ab,cd}$ is measured by forcing a current from contact a to b, then measuring the voltage between terminals d and c. The carrier concentration, n, and Hall mobility, μ_H, are obtained using the following equations:

$$n = -\frac{1}{qR_H} \tag{3.8}$$

$$\mu_H = \frac{R_H}{\rho} \tag{3.9}$$

Variable-field and low-temperature Hall measurements can assist in determining the energy level(s) of dopants and can reveal the effects of surface accumulation/depletion layers. A fixed-field (0.55 T) room temperature Hall measurement system was available for this project.

3.3.2. Current-Voltage (I-V) measurements

Current voltage (I/V) measurements provide a means to extract key electronic parameters of a device. In this project, I/V characteristics were measured at room temperature using a probe-station and a Keithley 2410 source-meter interfaced to a personal computer.

3.4. TCAD modelling

During the last decade, Technology Computer Aided Design (TCAD) software has been improved to better support the development and optimization of semiconductor materials, devices and circuits. It can provide deeper insight and understanding of physical processes involved in electronic device operation. According to an estimate from the International Technology Roadmap for Semiconductor (ITRS), the use of TCAD can reduce time required for device development by up to 37% [37].

In this thesis, Sentaurus TCAD tools from Synopsys have been used as a device simulator. A diagram describing basic implementations of TCAD simulation is shown in Figure 3.11. The simulation starts with creating device structures using either *Sentaurus Process* (*Sprocess*) or *Sentaurus Structure Editor* (*SDE*). The device structure is then divided into many nodes (mesh) using *Sentaurus Mesh* (*SMesh*) for applying finite-element analysis. Simulation continues with solving equations that describe the carrier transport in the active region of the

device using *Sentaurus Device* (*SDevice)*. The simulation results are further extracted, processed and visualized using *Sentaurus Visual* (*SVisual*).

Finite-element method (FEM) is a popular numerical method for solving engineering problems. FEM solves the problem by subdividing a large system into smaller, simpler parts that are called finite elements. The computations are performed at each node and then assembled together to model the entire device [38]. FEM has several advantages such as accurate representation of complex geometry of a device, inclusion of dissimilar material properties, easy representation of total solutions and capture of local effects [39]. Overall, to ensure the validity of simulation results a high granularity of nodes is required. However, as the mesh becomes denser, more equations need to be solved, resulting in a significant increase in computational time and power. Meshing optimisation needs to be performed to compromise between accuracy and time consumption of the simulation.

The output of the *SMesh* is used by *Sentaurus Device* (*SDevice*) to perform simulation on electrical and optical characteristics of the device. Such information as device dimensions, material/doping profiles, and location of electrical contacts are included in an input file for the program. The device simulation is based on Poisson's equation and current continuity equations coupling with a list of relevant semiconductor physics models. The simulation is controlled by a command file containing the details of physics models as well as conditions that need to be simulated. The equations to be solved are [40]:

The Poisson equation:

$$\nabla \cdot (\epsilon_S \nabla \psi) = -q(p - n + N_D - N_A) - \rho_{trap} \tag{3.10}$$

The electron and hole continuity equations:

$$\nabla \cdot J_n = qR_{net,n} + q\frac{\delta n}{\delta t} \tag{3.11}$$

$$-\nabla \cdot J_p = qR_{net,p} + q\frac{\delta p}{\delta t} \tag{3.12}$$

And the electron and hole current density equations:

$$J_n = q\mu_n nE + qD_n \nabla n \tag{3.13}$$

$$J_p = q\mu_p pE - qD_p \nabla p \tag{3.14}$$

Therein, ϵ_S is the dielectric constant of the semiconductor, ψ is the electrostatic potential, q is the electron charge, n and p are the electron and hole concentrations, N_D and N_A are the concentrations of ionized donor and acceptor impurities, ρ_{trap} is the surface charge density presenting interface trap charges (if any), J_n and J_p are the electron and hole current densities, R_{net} is the net recombination rate, E is the electric field, μ_n and μ_p are the electron and hole mobilities, and D_n and D_p are the electron and hole diffusion coefficients. The diffusion coefficients are related to the mobilities by Einstein relations:

$$D_n = \frac{\mu_n k_B T}{q} \tag{3.15}$$

$$D_p = \frac{\mu_p k_B T}{q} \tag{3.16}$$

where k_B is Boltzmann's constant and T is temperature.

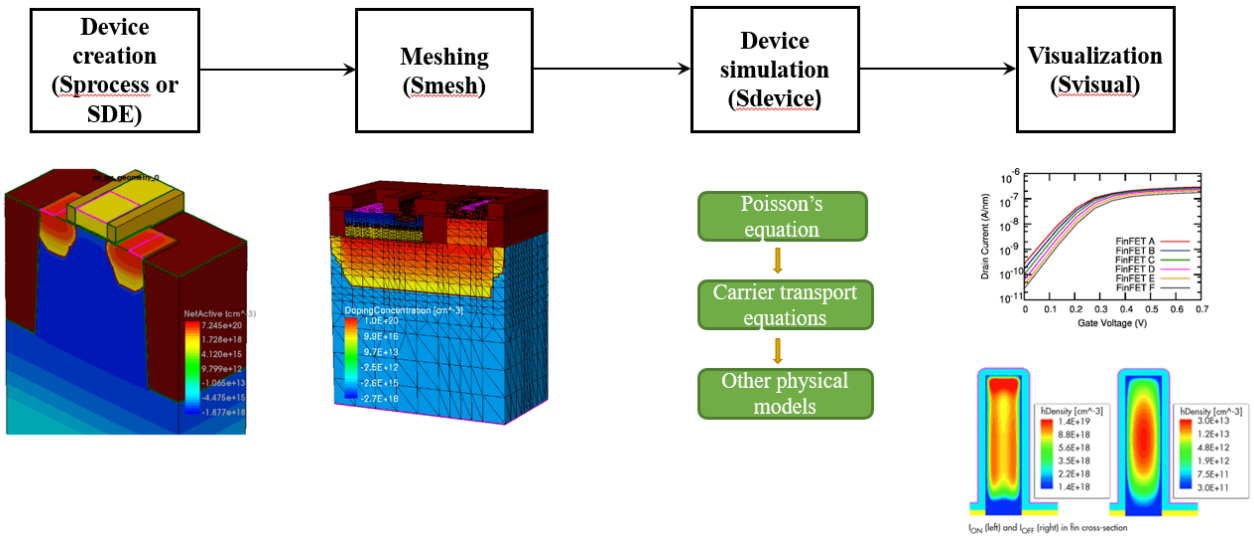


Figure 3.11: Schematic representation of the flow in Sentaurus TCAD simulation

The role of TCAD software in this project was as an aid to understanding the critical parameters in junction devices and how these parameters affected device characteristics. Since the standard TCAD library does not contain material files for emerging electronic materials such as a-C or a-ZTO, new material files were written using parameters from the literature. Simulations were then performed in accordance with the experimental data to determine the values to provide best fit models. Details of the simulation are discussed in the Chapter 4.

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Chapter 4

Energetic deposition, measurement and simulation of graphitic contacts to 6H-SiC

This chapter describes fabrication and simulation of graphitic contacts to 6H-SiC. Sentaurus TCAD is used to produce finite element models of the junctions and simulated I-V characteristics are compared with measurements. The effects of interfacial layers and varying work function are investigated using the models and methods for improved device performance are suggested.

4.1. Introduction

Since the discovery of carbon nanotubes and graphene, device applications for carbonaceous materials have been widely reported (see for example [1-5]). In particular, graphene and graphitic-carbon contacts to semiconducting materials have demonstrated excellent chemical/thermal stability, and Schottky barrier ideality factors close to 1.0 [6-8]. The latter is indicative of metal-semiconductor (MS) junctions with homogeneous interfaces.

Either by design or necessity, MS junctions occur in most electronic devices. The barriers to electron transport created at these junctions are to some degree determined by the choice of metal/semiconductor and by the fabrication techniques employed [9-10]. However, experimental

measurements of barrier height rarely correspond with theoretical predictions. This is due, in part, to interface layers/defects that are difficult to measure and/or simulate [9]. It is therefore desirable to develop models that describe the characteristics of these junctions and identify the most critical of the numerous parameters.

In this chapter, measurements and simulations of junctions between carbonaceous contacts (with differing microstructures and properties) and 6H-SiC are reported. With high breakdown field, electron saturation drift velocity and thermal conductivity, silicon carbide is an attractive wide bandgap semiconductor for high-power applications [11-12]. The formation, characterization and improvement of electrical contacts to all silicon carbide polymorphs has been of interest for some time (see for example [13-14]) and the aforementioned thermal stability of carbonaceous materials suggests applications for C/6H-SiC junctions as high-power rectifiers. Electrical and microstructural measurements have been coupled with finite element models (constructed using TCAD Sentaurus software [15]) to reveal the most critical parameters in these devices.

4.2. Experimental details

4.2.1. Device structure

The C contacts were deposited onto n-type 6H-SiC substrates (room-temperature carrier concentration, $n = 1.3 \times 10^{18}$ cm^{-3}) from the ionized C plasma produced by a Nanofilm Inc. filtered cathodic vacuum arc (FCVA) [16-17]. Prior to deposition, reverse side In contacts were formed on the 6H-SiC substrates to provide ohmic contact to the FCVA sample holder. Circular apertures of diameter 200 μm were then lithographically patterned in AZ1512 photoresist on the

front surface. Wet/dry etching to remove native oxide (eg. HF acid immersion) was not performed. The samples were mounted on the biased holder in the FCVA system and C (~50 nm in thickness) was deposited under previously reported conditions [16]. In the work described in [16], the energy of the depositing C fluxes was varied from 100 – 1000 V at room temperature and at elevated temperature (300 ºC). It was found that lower deposition energies resulted in resistive amorphous (sp^3 rich) carbon contacts. With higher average energies (> 500 eV) and moderately elevated temperatures, the sp^2 (graphitic) fraction in carbon films increased and rectifying contacts were formed between the carbon and 6H-SiC junctions. Here, deposition energies of 500 and 700 eV were selected based on the findings in [16]. These energies provided high sp^2 fraction, and therefore low through-film (contact) resistance with low enough landing energy to prevent damage to the substrate. Finally, Pt (~5 nm in thickness) was deposited onto the substrates to enhance optical contrast (for probing) before the Pt, C and photoresist coated 6H-SiC substrates were immersed in acetone/methanol to define the junctions (see Figure 4.1(a)). The current/voltage (I-V) characteristics were measured at room temperature using a probe-station and a source-meter interfaced to a personal computer.

The sp^2 fraction of the C films was determined using X-ray photoelectron spectroscopy, as described elsewhere [18]. Cross-sectioned devices were imaged in a JEOL 2100F transmission electron microscope (TEM) operating at 200 kV. The image in Figure 4.1(b) reveals the interface structure of the junction formed with the highest energy C flux (700 eV). The native oxide has been removed (sputtered away) by the C ions during deposition. Previous experiments have shown that sputtering is negligible at the lowest achievable energy (~ 25 eV) but increases as the deposition energy is increased. Electron energy loss spectra (not shown) suggested the presence of an amorphous, carbon-rich layer ~ 3 nm in thickness separating the graphitic region of the

anode from the 6H-SiC. This layer is thought to form due to shallow implantation/sputtering that occurs during energetic deposition.

4.2.2. Models and parameters used in the simulations

Two-dimensional simulations were performed using the device module in the Sentaurus TCAD software package. For a device with given doping profiles and geometry, the Sentaurus device module solves Poisson's equation and the current continuity equations [19]. Figure 4.2 shows the two model structures used in the simulations. In mixed mode simulation, the equations relating to separate devices are solved separately before a circuit netlist is defined to connect the devices. Solutions are then obtained for the entire circuit [20]. This approach has been used to model the C/6H-SiC devices. The model of the diode is divided into two equal area junctions since this enables lateral inhomogeneity (resulting in varying barrier height) to be simulated. The adjustable parameters for each diode are electrode work function, ϕ_m, interfacial layer (IL) material/thickness, d_i, oxide thickness, d_{ox}, and series resistance, R_s. The standard TCAD library did not contain suitable material files for the carbon device layers so new material files were written for the anodes and interlayer(s) using parameters from the literature [21-26] (shown in Table 4.1). The bandgap and electron affinity of amorphous carbon (a-C) were also taken from the literature [23-24]. Simulations were performed to determine the values of the adjustable parameters that provided the best fit with the measured *I-V* characteristics.

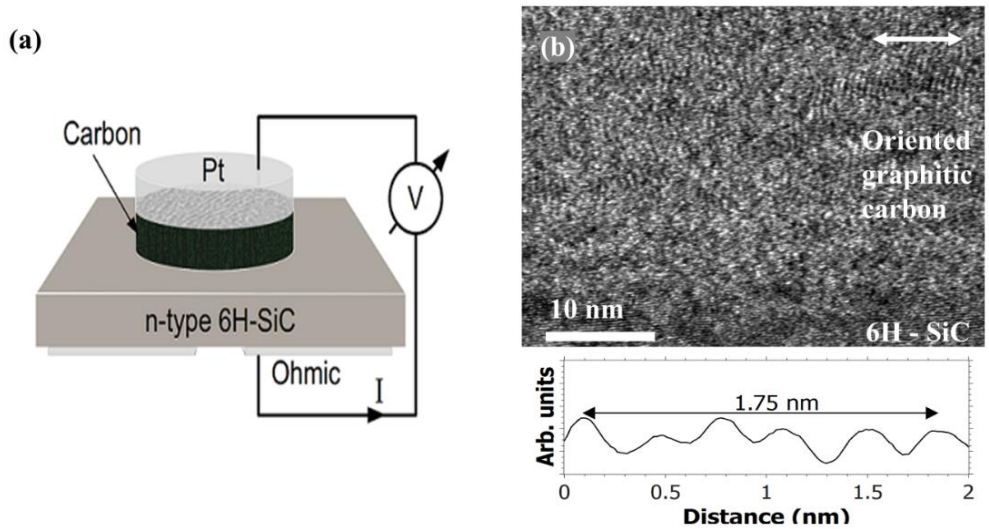


Figure 4.1: (a) Schematic and (b) cross-sectional TEM image of a junction between an oriented graphite anode and 6H-SiC substrate. In (b), the line profile is consistent with the characteristic inter-planar spacing of graphite and there is no native oxide layer at the C/6H-SiC interface.

Table 4.1: Material and electronic properties used in the TCAD models of C/SiC junction structures

Material Property	**6H-SiC**	**a-C**
Bandgap (E_g), eV	3.10	2.00
Electron Affinity (χ), eV	3.44	3.00
Density of States (N_c , N_v) at 300 K, cm^{-3}	6.82×10^{19}, 2.50×10^{19}	5.00×10^{18}, 1.80×10^{19}
Mobility (μ_c , μ_v), $cm^2/V.s$	420, 100	450, 380
Relative Permittivity (ϵ)	9.76	5.70
Thermal Conductivity (κ), W/K.cm	4.55	20.0

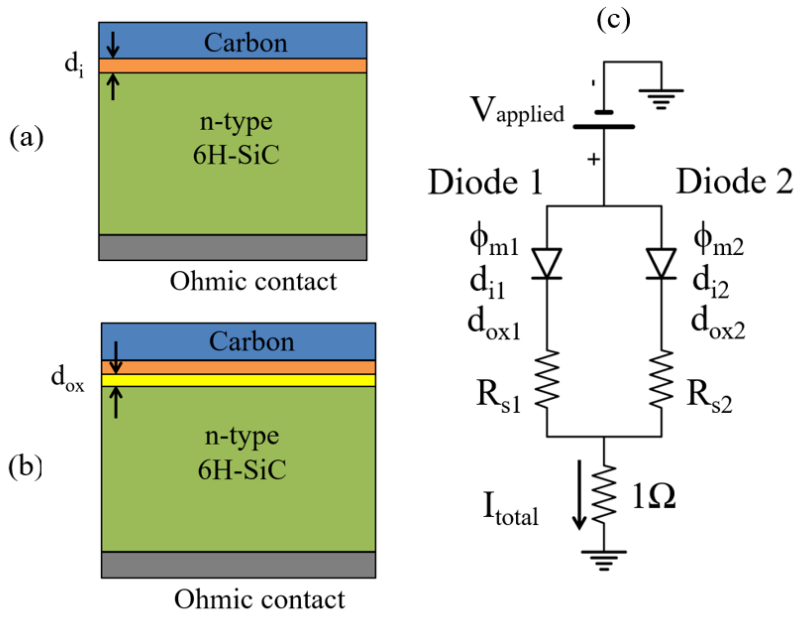


Figure 4.2: Structure of models used in simulations of C/6H-SiC junctions formed with C flux energies of (a) 700 eV and (b) 500 eV and (c) circuit diagram in mixed mode simulation

4.3. Results and Discussion

4.3.1. Device measurement

Typical experimental *I-V* characteristics from the C/6H-SiC contacts are shown in Figure 4.3. Lower energy carbon depositions (< 100 eV) resulted in the formation of highly resistive a-C contacts to the underlying 6H-SiC (as observed previously in C/Si junctions [27]). With increased deposition energy, the graphitic fraction of the contacts increased. This change was accompanied by increased non-linearity, forward current and rectification in the *I-V* characteristics.

According to thermionic emission (TE) theory, the *I-V* characteristic of a Schottky diode can be expressed by [10]:

$$I = I_s \exp(\frac{q(V-IR_s)}{nkT}) \tag{4.1}$$

where V is voltage across the diode, q is electron charge, k is Boltzmann's constant, R_s is series resistance, T is temperature, n is ideality factor, and I_s is saturation current, described by:

$$I_s = AA^*T^2 \exp\left(-\frac{q\phi_B}{kT}\right) \tag{4.2}$$

where A is contact area, A^* is the effective Richardson constant (~194 A/cm^{2}K^2 for 6H-SiC [28]) and ϕ_B is effective barrier height. From the measured forward bias *I-V* characteristics, n, ϕ_B and R_s were extracted using (4.1) and (4.2) (see Table 4.2). The rectification ratio (*RR*) is defined as the ratio of forward and reverse currents measured at ± 2.0 V. The contacts formed by carbon deposited onto substrates using deposition energy of 700 eV exhibited the highest rectification and ideality (*RR* of 10^4, I_s ~1.88 pA and n = 1.7). Lower deposition energy resulted in weakly rectifying junctions with larger series resistance.

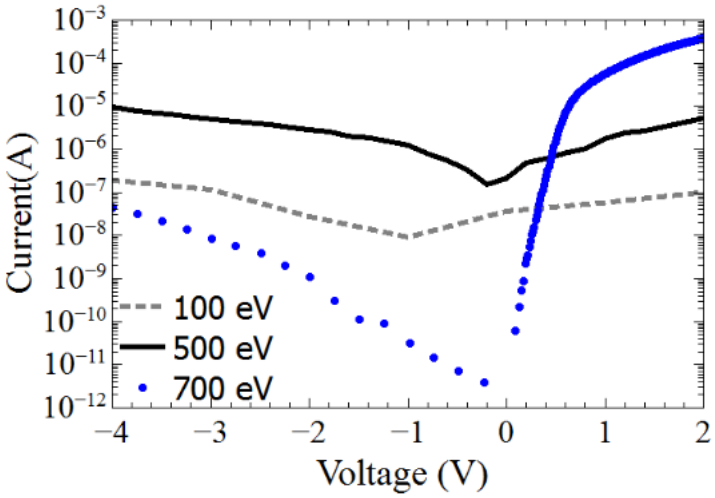


Figure 4.3: I-V characteristics of junctions between energetically deposited C and 6H-SiC for three deposition energies.

4.3.2. Models fitting to C/6H-SiC contacts

Figure 4.4 shows measured *I-V* characteristics from the C/6H-SiC junctions deposited at 700 eV (Figure 4.4(a) and (b)) and 500 eV (Figure 4.4(c)) with TCAD simulated *I-V* characteristics from the TCAD models. The heights of the potential barriers estimated from the simulated *I-V* characteristics of the respective best-fit models are 0.96 eV and 0.61 eV, comparable with those extracted directly from *I-V* measurements (Table 4.2). The models that provide best-fits to the measurement data are those with differing work-functions (indicated in Figure 4.2), supporting the assertions that the junctions are laterally inhomogeneous. Variations in the electrode work functions contribute to extracted ideality factors that exceed unity.

The structures of the TCAD models that provided best agreement between simulated and experimental *I-V* characteristics mimicked those of the fabricated devices (see Figure 4.1). More specifically, the best fit model of the C/6H-SiC interface at the junction formed from C

deposited at 700 eV is oxide-free (no IL) while the best-fit model of the C/6H-SiC junction formed from C deposited at 500 eV features an IL. As shown in Figure 4.4, this IL significantly affects the *I-V* characteristics of the junctions. Comparing the *I-V* characteristics in Figure 4.4(a) from the model/ experiment in which a 3 nm thickness a-C interface layer was present with those from a model with no a-C interface layer shows a reduction of three orders of magnitude. As previously reported, the thin insulating interface layer improves the *RR* of the junction by a similar amount since it causes little reduction in forward current [10]. Other values used in the models are presented in Table 4.3.

Table 4.2: Parameters extracted from measurements on C/6H-SiC. 90 % of the devices exhibited ideality factors within ± 10% of the values presented in the table.

Dep.energy	sp^2 %	I_s (nA)	n	R_S (Ω)	ϕ_B (eV)	RR
500 eV	80 ± 3	10.0 ± 0.2	> 5	67k ± 0.2k	0.65 ± 0.05	2
700 eV	85 ± 5	0.0018 ± 0.0005	1.7 ± 0.2	1.8k ± 0.1k	0.88 ± 0.05	10^4

4.3.3. Effect of work functions and interfacial thicknesses on the device performance

Once best-fit models were established, further simulations with these models were used to assess the sensitivity of the *I-V* characteristics to the work function of the graphitic electrode. Figure 4.4(b) shows the effect of varying ϕ_m. The models used here consisted of either dual parallel junctions (simulating an inhomogeneous junction) or a single junction. The best correspondence between experiment and simulation is provided by the dual junction model with ϕ_{m1} = 4.2 eV and ϕ_{m2} = 4.6 eV. The additional simulated curves show the effect of lateral inhomogeneity. The *RR* is highest with ϕ_m = 4.4 eV (no lateral inhomogeneity) and decreases

with decreasing lateral homogeneity. Figure 4.4(c) shows a similar plot for the C/6H-SiC diode formed with lower energy C. The trend is the same as in Figure 4.4(b) and the best correspondence is achieved with a similar dual junction exhibiting ϕ_{m1} = 3.8 eV and ϕ_{m2} = 4.4 eV. These lower ϕ_m values are consistent with the lower sp^2 content of the C contacts (Table 4.2) [29-30]. Also shown in Figure 4.4(b) is a simulated *I-V* characteristic from a laterally homogeneous junction with a higher ϕ_m (4.8 eV). A significantly higher *RR* is achieved. Br doping has been employed to increase ϕ_m in graphitic carbon [31] and in future work, Br will be introduced during the energetic deposition process to improve the electrical properties of these junctions.

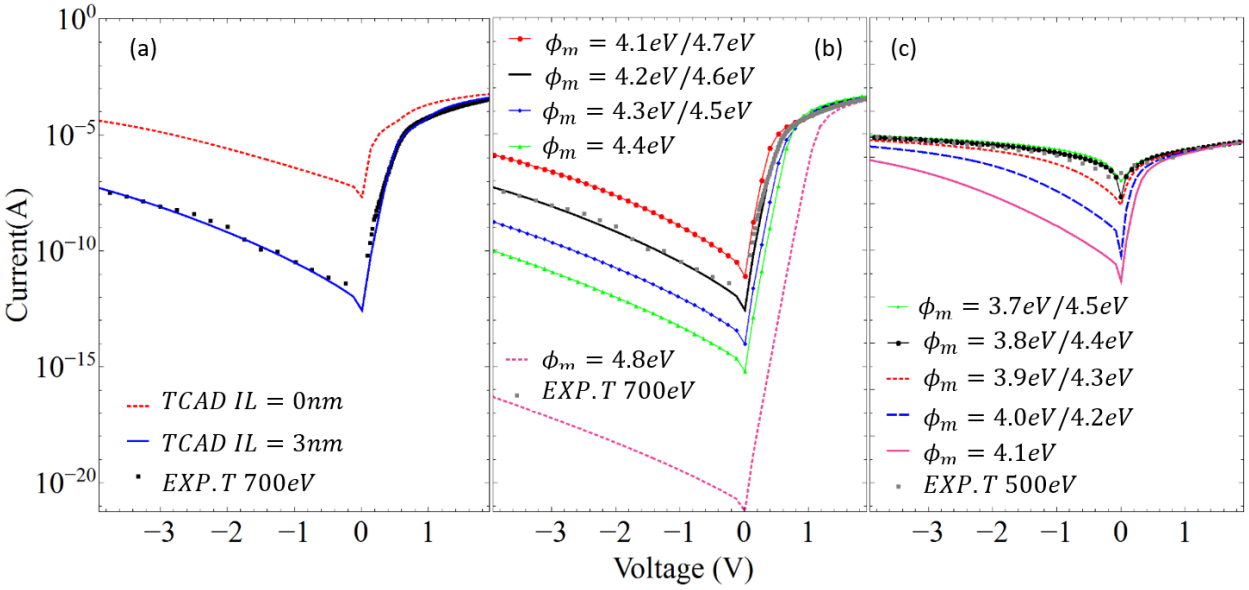


Figure 4.4: TCAD simulation results and experimental data of C/6H-SiC junctions showing (a) the effects of interfacial layer thickness and in (b) and (c) the effects of different ϕ_m and lateral homogeneity. Other model parameters are shown in Table 4.3.

Table 4.3: Parameters used in best-fit models of C/6H-SiC junctions. The series resistance (evaluated at higher forward bias) and the work functions derived from the simulations are broadly consistent with the data in Table 4.2, i.e. for the 700 eV diode, the experimental series resistance is 1.8 kΩ and the model series resistance is 2.3 kΩ, similarly, for the 500 eV diode, the experimental and model series resistance are 67 kΩ and 83 kΩ, respectively. The heights of the potential barriers estimated from the simulated I-V characteristics of the respective best-fit models are 0.96 eV and 0.61 eV, comparable with those extracted directly from I-V measurements shown in Table 4.2

	700 eV		**500 eV**	
	Diode 1	**Diode 2**	**Diode 1**	**Diode 2**
Electrode work function, ϕ_m (eV)	4.2	4.6	3.8	4.4
Series resistance R_s (Ω)	10k	3k	500k	100k
a-C IL thickness, d_i (nm)	3.0	3.0	3.0	3.0
a-C IL bandgap, E_{gi} (eV)	2.0	2.0	2.0	2.0
a-C IL electron affinity, χ_i (eV)	3.0	3.0	3.0	3.0
Oxide thickness, d_{ox} (nm)	0.0	0.0	0.0	2.2

4.4. Conclusion

Microstructural and electrical measurements have been performed to study graphitic rectifying junctions to n-type 6H-SiC. Finite element modeling (using Sentaurus TCAD) of these junctions elucidated the role of interface layers and electrode work function. The device simulations also suggest that improved rectification ratios may be obtained from C/6H-SiC Schottky junctions if their lateral homogeneity is improved and/or the thickness/properties of interfacial a-C/SiC mixing layers are more accurately controlled. This may be achieved by doping. In addition, whilst the TCAD simulations provided agreement with experiment using

parameters obtained directly from the literature, more detailed experimental work such as temperature dependent I-V measurements would enable more constraints to be imposed on the device models to provide more confidence in their accuracy.

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Chapter 5

Codeposition of amorphous zinc tin oxide using high power impulse magnetron sputtering: characterisation and doping

This chapter describes the deposition and characterisation of amorphous zinc tin oxide (ZTO). Thin film ZTO has been energetically deposited at 100 ºC using high power impulse magnetron sputtering (HiPIMS). The electrical and optical properties of the film are studied as a function of composition. This method produces high quality a-ZTO with low unintentional doping. Hydrogen was introduced after growth and resulted in stable moderate n-type carrier concentrations with no significant microstructural alterations. Electrical characterisation of these doped films is also included.

5.1. Introduction

Transparent conducting oxide (TCO) thin films are used in optoelectronic devices including flat panel displays, photovoltaics, electrochromic devices and anti-static coatings [1]. Due to its high carrier mobility (up to ~ 100 cm^2/Vs) and low visible-light absorption, indium tin oxide (ITO) is the most commonly used TCO [1]. However, In-free TCOs have been sought for some time due to the cost and relative scarcity of In. Among these, alloys of ZnO and SnO_2 (commonly referred to as ZTO) offer acceptable electron mobility (>10 cm^2/Vs), high thermodynamic stability and low processing temperatures [2-4].

Amorphous oxides composed of heavy-metal cations with (n-1) $d^{10}ns^0$ ($n \geq 4$) electronic configurations exhibit high electron mobility [5]. The large diameter and spherical symmetry of the ns^0 ($n \geq 4$) orbitals lead to significant overlap and conduction band dispersion [2,6]. In ZTO, Sn 4+ and Zn 2+ are such heavy-metal cations so amorphous ZTO (a-ZTO) exhibits comparable carrier mobility to its polycrystalline counterpart. This mobility is reported to be controlled by Sn 4+ rather than Zn 2+ because the $5s^0$ orbital of Sn 4+ is larger than the $4s^0$ orbital of Zn 2+ [7]. By comparison, carrier mobility in a-Si:H is limited because transport occurs by hopping between localized tail-states [8-10]. Provided suitable doping processes are available, applications for a-ZTO include transparent interconnects and active layers in transparent field effect transistors [2,7].

Low energy codeposition from Zn and Sn sources has enabled the properties of the resulting ZTO films to be determined as a function of composition and bonding [3,4,11]. For example, n-type carrier mobility has been found to peak in a-ZTO with Zn:Sn ratios of ~ 25:75 and 75:25 [3]. Previous reports describing energetic deposition of ZTO and other binary/ternary metal oxides suggest that growth from higher energy fluxes can enhance the structural, electrical and optical properties [12-15]. In particular, higher density and epitaxial growth have been observed [14,15] and attributed to dynamic annealing caused by thermal spikes from energetic incident ions [16]. The importance of high density in a-ZTO was demonstrated by Lee et al. who performed post-deposition processing to densify the channels of a-ZTO thin film transistors (TFTs) (deposited using low energy RF sputtering) and reported significantly improved channel mobility afterwards [17]. Energetic deposition also enables access to a greater variety of microstructures at lower growth temperatures and shallow implantation [18]. This is an attractive aspect for materials such as a-ZTO with possible applications including active device layers on

flexible plastics.

Here, structural, electrical and optical characteristics of a-ZTO thin films energetically deposited using high power impulse magnetron sputtering (HiPIMS) at a moderate growth temperature are presented. HiPIMS is a scalable deposition method enabling films to be deposited over large areas using standard magnetron sputtering guns with no harmful precursor/by-product gases required/generated. The DC or RF power supply is simply replaced with a pulsed high power supply. A low duty cycle ensures that the average power is comparable with conventional DC/RF sputtering and target over-heating is thus avoided. High quality metal oxide films grown by HiPIMS have demonstrated properties suitable for optical and electronic devices [15,19-21].

5.2. HiPIMS deposition of a-ZTO films

The ZTO film was codeposited at 100 ºC onto a-plane sapphire (4-inch wafer) using an AJA 4-gun magnetron sputter deposition system and RUP-7 pulsed power supply. Prior to deposition, the system was pumped to a base pressure $< 1 \times 10^{-6}$ Torr. The substrate was at a floating potential (~ 10 V) during the 20-min deposition. The Zn target (3-inch diameter, 99.99 % purity) was operated in HiPIMS mode (with pulse amplitude, width and frequency of 500 V, 50 μs and 200 Hz) whilst the Sn target (3-inch diameter, 99.99 % purity) was operated in direct current magnetron sputtering (DCMS) mode at 64 W. The 1:3 O_2:Ar process pressure was 6.0 mTorr. No instabilities occurred during the 20-min reactive deposition process.

The target voltage and current waveforms are shown in Figure 5.1. The RUP7 delivers a current while maintaining the pre-set output voltage, and thus has an output inductance. This

output inductance and small output capacitance causes the ringing at the start of the voltage pulse prior to plasma ignition at 10 µs. This delay in current onset is typical for HiPIMS systems and is well described by Anders [22]. After plasma ignition, the target current increases almost linearly until the end of the voltage pulse. This is again typical and caused by positive feedback leading to increases in plasma density due to ionization of sputtered flux and subsequent return of a proportion of that ionized flux to the target, where it sputters more material.

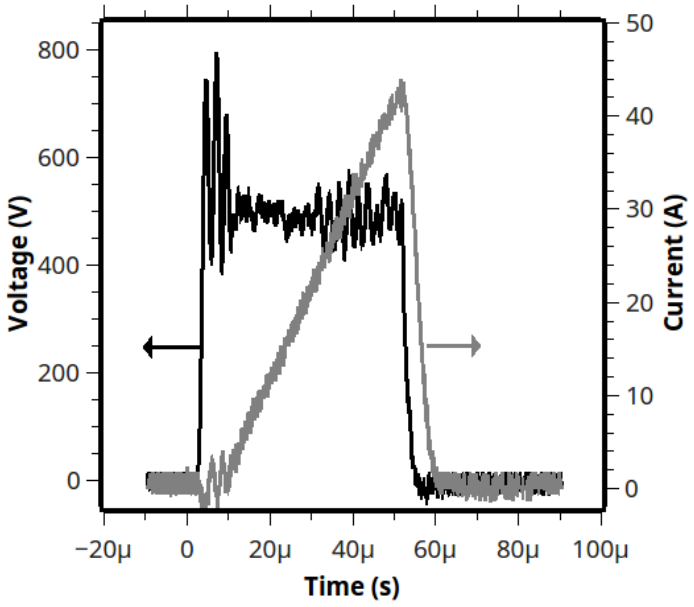


Figure 5.1: The HiPIMS voltage and current waveforms used to energetically deposit Zn

5.3. Film processing and characterisation methods

The film thickness, measured after growth using an atomic force microscope, varied between 70 nm at the Zn-rich side to 72 nm at the Sn-rich side. After deposition, the wafer/film was diced into 10 × 10 mm^2 samples. Five samples that spanned the compositional range were then selected for characterisation. X-ray diffractograms were collected using a Bruker D4

diffractometer with Cu K-α (8.048 keV) source and Bragg-Brentano configuration. Cross-sectioned samples were extracted and thinned to electron transparency using a FEI SCIOS focused ion beam system before being imaged in a JEOL 2100F TEM operated at 200 kV. Crystallisation of the ZTO under the electron beam was not observed during imaging or diffraction analysis.

The Zn:Sn ratio was determined by X-ray photoelectron spectroscopy (XPS) performed using a K-Alpha XPS system with an Al K-α (1487 eV) source. Surface contaminants were removed from the films within the XPS system using successive 8 s, 200 eV, Ar^+ plasma etches performed at a pressure of less than 10^{-8} Torr [12]. The O 1s, C 1s, Sn 3d, Zn 2p, and valence band spectra were collected from each sample before and after this surface cleaning process to study surface electronic properties.

Room-temperature Hall measurements were performed on the diced samples using an Ecopia HMS3000 Hall effect measurement system (with fixed field of 0.55 T). Thermally evaporated Al Van der Pauw corner contacts were deposited for these measurements and verified as Ohmic. Optical transmission measurements were performed using a Cary60 spectrophotometer. Optical band gaps were then determined from this data using Tauc's method [23].

Hydrogen annealing was performed on the samples in a vacuum system pumped to $<1 \times 10^{-6}$ Torr. Each diced sample was mounted on a substrate heater and heated up to 500 ºC in 20 mins. Hydrogen gas was then introduced to a pressure of 100 mTorr. The temperature and pressure were maintained for one hour before the sample was cooled to room temperature at 100 mTorr. The annealed samples were structurally, optically and electrically characterised for

comparison with the as-deposited samples.

5.4. Results and discussion

5.4.1. Structural characterisation of a-ZTO films

The compositional range achieved in the codeposited HiPIMS/DCMS film was Zn:Sn = 25:75 – 54:46. In this work, the samples are labelled according to their fractional Sn content eg. *ZTO 75* has Zn:Sn of 25:75. Figure 5.2 shows representative atomic force micrographs of the ZTO film at the (Figure 5.2(a)) Zn-rich and (Figure 5.2(b)) Sn-rich end-points. The root mean square (RMS) roughness of the film between these end-points was < 1 % of the film thickness. No measurable alterations in the surface morphology occurred during annealing.

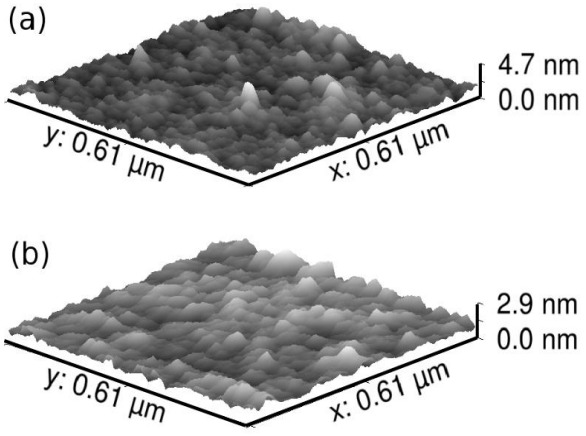


Figure 5.2: Atomic force micrographs showing the surface morphology of (a) *ZTO 75* and (b) *ZTO 46* as-deposited.

X-ray diffractograms (not shown) were collected from the samples before and after annealing. All were consistent with a-ZTO. Jayaraj et al. [24] previously reported that

crystallisation of their pulsed laser deposited a-ZTO films occurred at 450 ºC but no evidence of this was seen here. The higher temperature required for crystallisation is attributed to higher density caused by the energetic flux. The TEM image shown in Figure 5.3 is taken from *ZTO 69* (as-deposited). The image and the inset diffraction pattern, generated using a fast Fourier transform, are consistent with an amorphous structure and are therefore in agreement with the X-ray diffractogram collected from this sample.

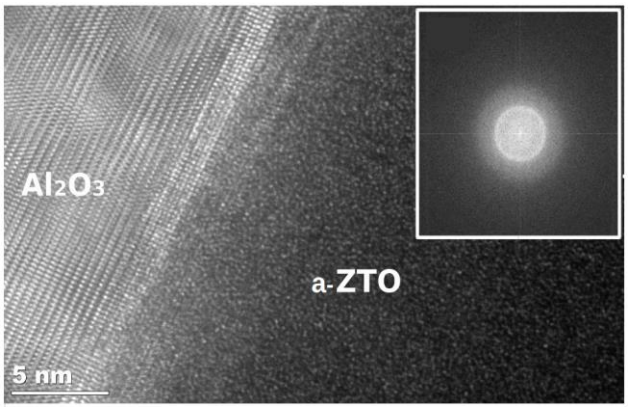


Figure 5.3: TEM image and (inset) diffraction pattern collected from a cross-sectioned as-deposited sample (*ZTO 69*).

5.4.2. Compositional analysis of thin films

The surface chemistry of the a-ZTO films was investigated in more detail using XPS measurements. Figure 5.4(a) shows a representative O 1s spectrum from *ZTO 69* (as-deposited). In addition to the peak attributable to lattice bonded O (at 531 eV), a strong higher energy peak attributable to surface-bound OH was produced. After etching within the XPS system for 30 s using 200 eV Ar ions, this OH peak was significantly reduced (Figure 5.4(b)). A valence band spectrum from the same as-deposited sample is shown in Figure 5.5. The valence band maximum (VBM) is well defined (by Sn(IV) bonding) with no significant features appearing in the band

gap. This was true of valence band spectra collected from all samples. In previous work, a-ZTO deposited energetically from a cathodic arc produced clearly identifiable acceptor state peaks (due to Sn(II) bonding) above the VBM [12]. The absence of these features in the HiPIMS a-ZTO film suggests the semi-insulating nature of the as-deposited film is not due to acceptor compensation of donors but due to low unintentional doping ie. the film is dense, high-purity a-ZTO.

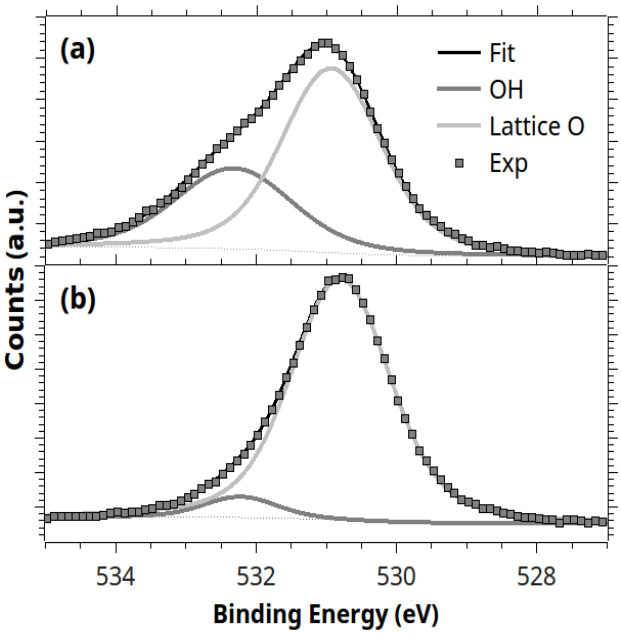


Figure 5.4: O 1 s spectra from ZTO (a) before and (b) after etching in the XPS system using 200 eV Ar ions

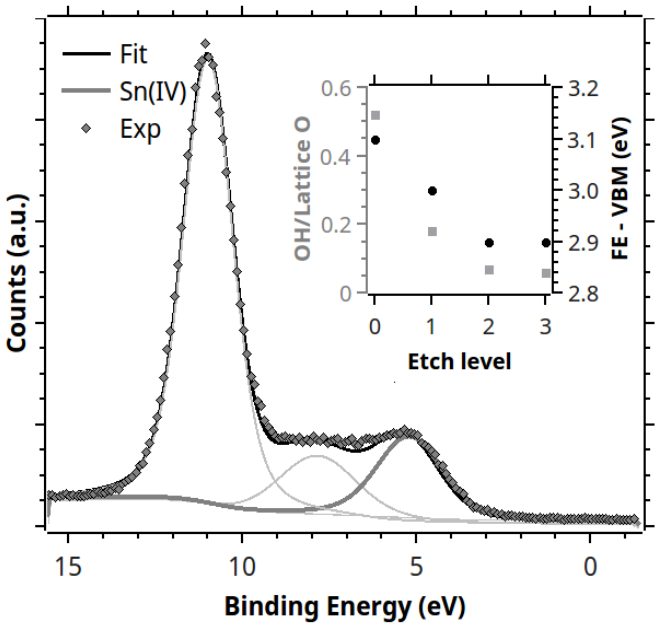


Figure 5.5: Valence band spectra from HiPIMS a-ZTO with fitted peaks and (inset) a plot showing the energy difference VBM – Fermi energy (FE) as a function of surface OH coverage (see text).

The energy difference between the VBM and the (0 eV) Fermi energy (FE), can be extracted from valence band spectra by extrapolating the valence band edge to the instrument background, as reported previously [25]. The Figure 5.5 inset shows that before etching, the energy difference VBM-FE exceeds the optical band gap in the as-prepared (OH-attached) a-ZTO, meaning the valence band is bent downwards at the surface. This downward band bending is accompanied by surface electron accumulation in n-type semiconducting materials including a-ZTO [26-27]. After the first etch, some of the OH is removed and the valence band maximum shifts 0.1 eV towards the Fermi energy. A second identical etch process results in further removal of the OH and an additional 0.1 eV shift. Similar energy shifts were observed in the O 1s and C 1s peaks. Much smaller changes in the O 1s and VBM energies are observed after additional (identical) etching, likely due to re-hydroxylation in the XPS chamber. The sputtering

was performed with minimal energy/time and intended to remove OH and little else. The recovery of the OH and reproduction of XPS valence band spectra following atmospheric exposure suggests the depth of sputtering was sub-nanometre. These measurements highlight the importance of surface preparation when forming electrical contacts to ZnO-based materials including a-ZTO.

5.4.3. Optical measurements

Figure 5.6(a) shows the Ultra Violet (UV)-visible transmission through the as-deposited a-ZTO film as a function of Zn:Sn. The inset shows a Tauc plot with data from the Zn-rich (*ZTO 44*) and Sn-rich (*ZTO 75*) samples. After annealing (Figure 5.6(b)), the visible transmission remained similar but the optical bandgap of all samples increased by approximately 0.1 eV (see Table 5.1). Due to the absence of microstructural changes observed using AFM, TEM and XRD, the widening of the bandgap following H_2 annealing is attributed to Burstein-Moss shifts [28]. These are likely to occur in ZTO due to the dispersed energy states forming the bottom of its conduction band [29].

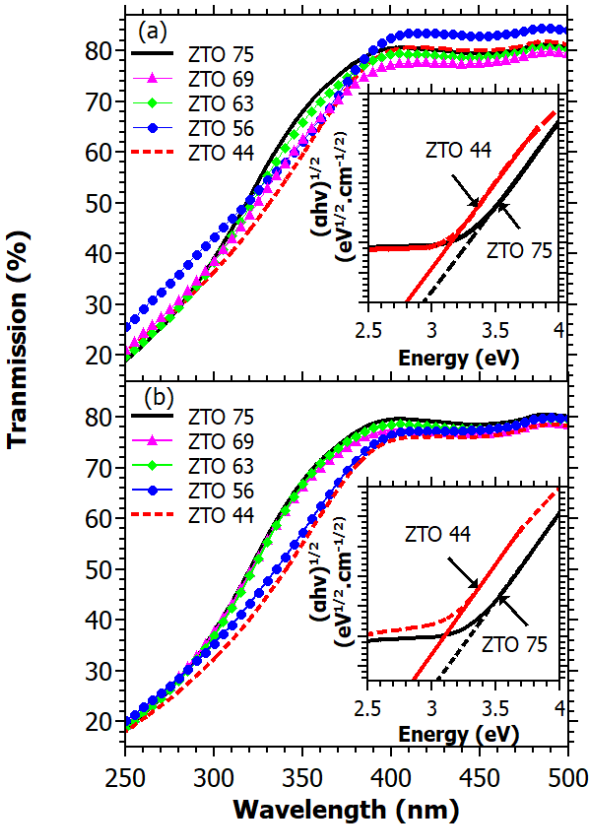


Figure 5.6: Optical transmission through (a) as-deposited and (b) annealed a-ZTO samples with differing Zn:Sn ratios. Tauc plots [23] taken from the Zn- and Sn- rich end points (*ZTO 44* and *ZTO 75*) are shown as insets.

Table 5.1 summarizes the structural, optical and electronic characteristics of the a-ZTO film as a function of Zn:Sn. As discussed earlier, the codeposited film was uniformly smooth and amorphous after growth. Unintentional doping was low and the film was semi-insulating which prevented Hall effect measurements. Across the compositional range, the film exhibited an indirect optical bandgap which varied little with composition. There is ongoing debate regarding the direct/indirect bandgap of a-ZTO. Although Cheng et al. [30] reported a direct bandgap for their a-ZTO films, numerous authors reported an indirect gap of 2.80 – 2.85 eV [24,31] in agreement with the values reported in the thesis. Importantly, the major findings in Chapter 5

(reduction of downward surface band-bending with OH removal) remain unchanged whether the gap is direct or indirect.

5.4.4. Hydrogen annealing of the HiPIMS a-ZTO films

The inherent, stable, n-type doping consistently observed in ZnO is widely believed to be caused by hydrogen bound within oxygen vacancy sites (H_o) [32]. Increased temperature and/or oxygen pressure during growth or post-deposition annealing causes the density of H_o to decrease since the density of oxygen vacancies decreases [32]. This trend has been observed in polycrystalline ZnO when deposited/annealed in oxygen ambient [13,33]. Similarly, n-type doping results from H_o in SnO_2 [34].

The a-ZTO were annealed in the presence of hydrogen to explore the possibility of doping and assess structural, optical and electrical properties post-annealing. Following a 1-hr 500 ºC anneal in 100 mTorr H_2, all samples remained amorphous but exhibited significantly reduced resistivity (ρ), moderate n-type carrier concentrations (n) and n-type carrier mobility (μ) ranging from 4.0 to 13 cm^2/V.s. The optical transmission of the films changed little apart from the aforementioned Burstein-Moss shifts [28]. The carrier mobility peaks twice across the compositional range. This is in agreement with previous findings [3]. However, numerous studies have also reported variations in mobility of a-ZTO with composition. In a study of the effects of channel stoichiometry on the electrical characteristics of ZTO thin-film transistors [35], the highest electronic mobilities were achieved with Zn:Sn ratio = 1:1. Another group found peak mobilities with Zn:Sn ratio at 1:3 and 4:1 for their ZTO films [3]. No firm conclusion regarding this broad variation has been stated. Several effects leading to this variation could be considered such as formation of sub-gap defect states [36], the nature of transition from

amorphous to crystalline phase which has been shown to be affected by thermal annealing [37], and so on. Further work needs to be conducted to fully understand the mechanisms underlying this variation. The electrical properties of the doped a-ZTO remained stable after a further 1 hr, 200 ºC anneal in air. These results show that annealing in hydrogen is an effective way to introduce carriers into a-ZTO. Device applications for the n-doped HiPIMS a-ZTO will be explored in future work.

Table 5.1: Summary of electronic and optical properties of HiPIMS a-ZTO. As-dep. and An. refer to measurements from as-deposited and annealed samples, respectively

	Optical gap (eV)		***r*** **(Ω.cm)**		***n*** **(10^{17} cm^{-3})**		***μ*** **($cm^2V^{-1}.s^{-1}$)**	
Zn:Sn	As-dep.	An.	As-dep.	An.	As-dep.	An.	As-dep.	An.
25:75	2.90 ± 0.05	3.00 ± 0.05	>1k	4.5 ± 0.5	-	3.5± 0.5	-	4.0 ± 2.0
31:69	2.80 ± 0.05	2.90 ± 0.05	>1k	3.7 ± 0.5	-	1.3 ± 0.5	-	13.0 ± 2.0
37:63	2.90 ± 0.05	3.00 ± 0.05	>1k	4.5 ± 0.5	-	2.8 ± 0.5	-	5.0 ± 2.0
44:56	2.80 ± 0.05	2.90 ± 0.05	>1k	6.9 ± 0.5	-	1.3 ± 0.5	-	7.0 ± 2.0
56:44	2.80 ± 0.05	2.90 ± 0.05	>1k	1.3 ± 0.5	-	4.2 ± 0.5	-	12.0 ± 2.0

5.5. Conclusion

HiPIMS has been employed to energetically and reactively co-deposit a thin film of amorphous n-type zinc tin oxide with a compositional (Zn:Sn) gradient. The microstructural, electrical and optical properties of the film were found to vary little with Zn:Sn. As-deposited,

the films were smooth, amorphous, transparent and semi-insulating. The introduction of hydrogen by post-deposition annealing resulted in significantly increased conductivity with moderate carrier concentrations (~ 10^{17} cm^{-3}) and n-type carrier mobility of 4 - 13 $cm^2/V.s$. These characteristics were achieved with no measurable crystallisation and proved stable. X-ray photo-electron spectroscopy (XPS) revealed that surface-bound OH could be removed by light Ar etching to alleviate downward surface band-bending in the annealed a-ZTO. The optical and electronic properties of the HiPIMS a-ZTO coupled with the low roughness, low optical absorption and response to hydrogen annealing suggest suitability as a material for transparent electronic devices.

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Chapter 6

Observation and characterisation of memristive silver filaments in amorphous zinc tin oxide

Lateral devices consisting of planar Ag electrodes (with separation 200-700 nm) supported on thin films of amorphous zinc-tin-oxide (a-ZTO) have been constructed to study Ag migration in a-ZTO and characterised to assess the suitability of these materials for memristive devices. After a high-field forming process, these simple devices exhibited volatile, bidirectional resistive switching. Detailed electrical and structural measurements from these devices are presented in this chapter and show that they are capable of emulating biological synaptic behaviour.

6.1. Introduction

Amorphous oxide semiconductors (AOS) can be deposited over large areas with uniform structural and electrical properties [1]. These materials have received considerable research and industry attention, largely due to optical and electronic properties that make them suitable for transparent thin film transistors (TFTs), a building block for many types of display and interactive electronics [1]. Their flexibility and applicability to wearable electronics has also been demonstrated by numerous research groups [1,2].

The application of AOS in devices other than transistors, notably memristors and resistive random access memory (RRAM), is gaining interest. For these devices, the oxide is typically inserted between metallic electrodes to form metal–insulator–metal (MIM) structures, a two-terminal architecture well suited to implementation within three-dimensional arrays. Combining memristors and transistors for advanced neuromorphic circuits is an exciting prospect and self-programming capabilities have already been demonstrated using Si-based devices [3]. If these devices and circuits were based on AOS, then transparency, simplified fabrication and/or flexibility would be added benefits.

Indium-gallium-zinc-oxide (IGZO) is currently the dominant transparent AOS for TFTs but amorphous zinc-tin-oxide (a-ZTO) is a low-cost indium-free alternative. Thin films of a-ZTO have been deposited using numerous methods including chemical vapour deposition [4], conventional physical vapour deposition (PVD) [5] and plasma assisted PVD [6]. The latter has produced metal oxide layers of sufficient quality for transparent ZnO transistors [7,8], high-K dielectric layers [9] and more recently, a-ZTO for devices [10,11]. Previously, it was shown that combined DC and high power impulse magnetron sputtering (HiPIMS) can produce a-ZTO at moderate growth temperatures with very low unintentional doping and high carrier mobility [10]. Furthermore, the n-type carrier concentration in this material can be altered controllably by annealing the films in the presence of hydrogen [10]. As with other AOS, much of the device research performed on a-ZTO has involved thin film transistors (see for example [4,5]). It has been shown that partially oxidised Ag (AgO_x) can form highly rectifying contacts to a-ZTO and that these contacts, when employed as gates in metal-semiconductor field-effect-transistors (MESFET) devices, exhibit high stability under normal operating voltages/fields [4]. However,

in ZnO MESFETs with AgO_x gates, bias stress induced degradation has been observed and attributed to Ag migration/diffusion into the channel region [12].

Electric-field driven Ag migration has been reported in oxides, oxynitrides and carbon-based materials (see for example [13,14,15]). Characteristics observed include emulation of the Ca^{2+} or Na^{+} dynamics of a biosynapse in diffusive Ag-in-oxide memristors with Pt/SiO_xN_y:Ag/Pt structure [15]. Despite reports of similar behaviour in AgO_x/a-ZTO [4] Schottky devices, memristive behaviour in these material combinations has received comparatively little attention. Resistive switching has been demonstrated in IGZO [16] and a-ZTO [17,18] and resistive switching and emulation of learning functions in annealed a-ZTO/Ag junctions were recently reported [11]. In the latter, electrical characterisation suggested the underlying mechanism but direct observation of this mechanism was not possible due to the vertical device architecture. Here, a lateral architecture has been adopted for the devices which has enabled microscopy of the region between the contacts before and after electrical characterisation. The existence (or absence) of a filament within the devices can be verified in this way and the underlying switching mechanisms can be better understood.

6.2. Methods

The a-ZTO film was co-deposited at 100 °C onto SiO_2 and Al_2O_3 using an AJA 4-gun magnetron sputter deposition system and RUP-7 pulsed power supply. Prior to deposition, the system pressure was less than 1×10^{-6} Torr. The potential of the substrate floated (at ~ 10 V) during the 20 min deposition. The Zn and Sn targets were both 3-inch diameter, 99.99% purity and they were respectively operated in HiPIMS mode (with pulse amplitude, width and frequency of 600 V, 150 μs and 100 Hz) and in DC magnetron sputtering mode at 75 W. The 1:3

O_2:Ar process pressure was 6.0 mTorr. No instabilities occurred during the 20 min. reactive deposition process. The resulting film thickness was 70 nm. Further details of similar HiPIMS a-ZTO deposition processes as well as target current/voltage waveforms are reported elsewhere [10]. Following deposition, the wafers were diced into 10 × 10 mm^2 substrates for optical, structural and electrical characterisation followed by device fabrication. X-ray photoelectron spectroscopy (in a K-alpha system with monochromated Al K-α source) was used to determine the composition. An Asylum Research MFP-3D Infinity atomic force microscope (AFM) was used to assess surface topography/roughness and measure the film thickness from step-height measurements.

UV-Visible spectroscopy was performed using a Cary 50 UV-Vis spectrometer to determine transparency and optical band-gap. Spectroscopic ellipsometry was performed using a J. A. Woolam M-2000 to determine the refractive index of the a-ZTO. X-ray diffraction using a Bruker D4 in Bragg-Brentano configuration confirmed the amorphous microstructure of the films. Al Van der Pauw contacts were thermally evaporated and electrical measurements using an Ecopia HMS-3000 system revealed the as-deposited 10 × 10 mm^2 films to be semi-insulating.

Electron beam lithography was used to pattern PMMA electron-beam resist (950 PMMA A4, thickness 200 nm). A FEI Nova NanoSEM interfaced with Nabity NPGS electron beam writing software was used for this purpose. The operating conditions were: 20.00 kV accelerating voltage, working distance 5 mm, beam current 18 pA. A Gatan PECS ion-beam deposition system was then used for the (1st) Ag and (2nd) Pt device layers (each 30 nm in thickness). The Pt capping layer assisted in providing good contact to the device during probing. An image of the completed device and a schematic of the cross-section/measurement circuit are shown in Figure 6.1. Out-diffusion of oxygen from the a-ZTO device layer is expected to yield a

substoichiometric silver-oxide interface layer. This out-diffusion has been observed in similar devices using transmission electron microscopy and electron energy loss spectroscopy [19].

Room-temperature two-probe current-voltage characteristics were measured on a Keysight twin channel source-measurement unit with the device mounted on a probe-station. Finally, a FEI SCIOS electron microscope was used to image the pristine and switched devices. The operating conditions were; 5.00 kV accelerating voltage, working distance 2.9 mm, beam current 200 pA and substrate bias -4.00 kV.

6.3. Resistive switching performance

The resistivity of the a-ZTO film (with composition Zn:Sn = 44:56) exceeded 10 kΩ.cm. Optical transmission through the 70 nm film exceeded 80% across the visible spectrum and the optical band-gap was 2.9 eV. As-deposited, the device exhibited high resistance at low bias, as shown in the 'forming' characteristics of the device in Figure 6.1(d). A current limit of 100 nA was imposed during this voltage sweep. Figure 6.1(c) shows I-V characteristics from a device with identical geometry to that shown in Figure 6.1(a) and patterned on the same a-ZTO film but without the Ag device layer. As shown, this Pt contacted device exhibited I-V hysteresis, suggesting that mobile defects moved under the action of the applied E-field. The markedly different I-V characteristics in Figure 6.1(d) come from the device shown in Figure 6.1(a) with the Ag device layer. At low bias (below the threshold voltage), the I-V characteristic is approximately linear (see inset Figure 6.1(d)). This is consistent with the characteristics of back-to-back (metal-semiconductor-metal) Schottky diodes, in which emission over the potential barriers is excluded and the dominant carrier transport mechanism is direct tunneling through the potential barriers [20]. With an applied electric field of 0.5 MV/cm, the current (and

conductance) increases sharply. This electric field is lower than the 2-4 MV/cm field reported to form a filament in a vertical Al/a-ZTO/Pt structure [17]. Ag is known to diffuse more rapidly in ZnO (with rates of ~ 1×10^6 cm^2/s quoted) than it does in many other oxides used in Ag-filament based memristors, including TiO_2 [21]. However, the relative ion mobilities of Ag and Al in a-ZTO of the composition employed here are unknown.

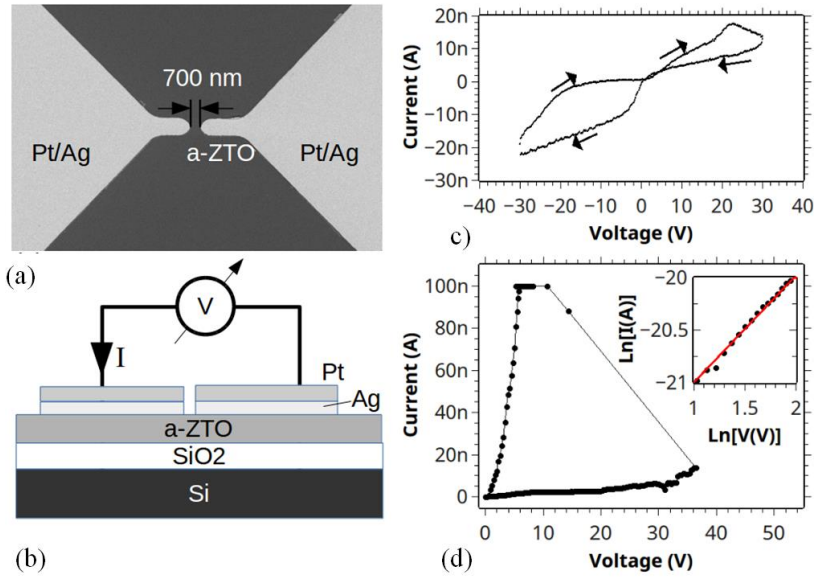


Figure 6.1(a) Electron micrograph and (b) cross-sectional schematic of the Pt capped Ag/a-ZTO memristor device. I-V characteristics taken from devices with (c) Pt (only) electrodes and (d) with Ag/Pt electrodes. At low bias in the pristine device state, the slope of the Ln(I) versus Ln(V) plot (inset in (d)) is unity, consistent with transport by direct tunneling.

Figure 6.2(a) shows I-V characteristics measured immediately after the forming step in Figure 6.1(d). These measurements, taken with a 100 nA current limit imposed and with one complete sweep taking 18 s, show the pinched hysteresis loop typical of memristors. The switching between the high- and low- resistance states (HRS and LRS) occurs above a threshold for both positive and negative voltages. Figure 6.2(b) shows the cumulative probability of a

switching event as a function of threshold voltage. This plot is approximately symmetric, as expected when the two electrodes are composed of identical materials. The positive and negative threshold voltages reduced with increased operation, as shown in Figure 6.2(b) where selected points are labelled with their respective sweep numbers. The cumulative probability versus threshold voltage characteristic can be fitted using the Weibull function [22,23]:

$$F(x) = 1 - \exp\left[\left(-\frac{x}{x_o}\right)^{\beta}\right] \tag{6.1}$$

where F is the switching probability, x is a positive/negative voltage, x_o is the average value of $V_{threshold}$ and β is the Weibull slope. The curves on both sides of the origin are fitted well by single values of β, indicating a single conducting pathway (filament) [23]. At voltages below the switching threshold in the HRS, the I-V relationship can be fitted with the Schottky emission equation (see Figure 6.2(c)) as in the work of Murali et al. [17]. Using this equation, a slope of 0.91 and a contact separation of 700 nm, yields a relative dielectric permittivity of 3.7. Following the approach of Murali et al. [17], the high frequency dielectric permittivity was obtained from the square of the refractive index of the a-ZTO film, measured using spectroscopic ellipsometry. This measurement yielded a relative dielectric permittivity of 3.6. Poole-Frenkel (P-F) emission was excluded as the dominant transport mechanism since the P-F equation provides a less accurate fit to the data and fails to account for asymmetry observed in the low-bias I-V characteristics. Furthermore, the dielectric constant required for best correspondence to the data exceeded any published values for a-ZTO [17,18].

6.4. Mechanism of device operation

As the current falls below the current limit of 100 nA in the LRS, the I-V characteristics appear to remain consistent with transport by Schottky emission (Figure 6.2(c)). If the relative permittivity of the a-ZTO between the electrodes is assumed constant, a new effective contact separation less than $1/10^{th}$ the separation between the electrodes can be calculated from this LRS slope. An incomplete filament between the electrodes, containing for example gaps and/or inter-granular barriers, could produce this result. Potential barriers existing at the Ag electrodes in the form of AgO_x interface layers are also likely. Conductance in the LRS at lower voltages (dashed box in Figure 6.2(c)) appears to be limited by Schottky barrier(s) since this region is linear on a semi-logarithmic I-V plot (Figure 6.2(d)). This characteristic is attributed to Schottky barrier(s) at the electrodes and/or at gaps in the filament. After 50 sweeps with a 100 nA current limit imposed, the I-V relationship in the LRS suggested multiple current limiting mechanisms with an unstable filament leading to volatile behaviour. Ohmic conduction was not observed.

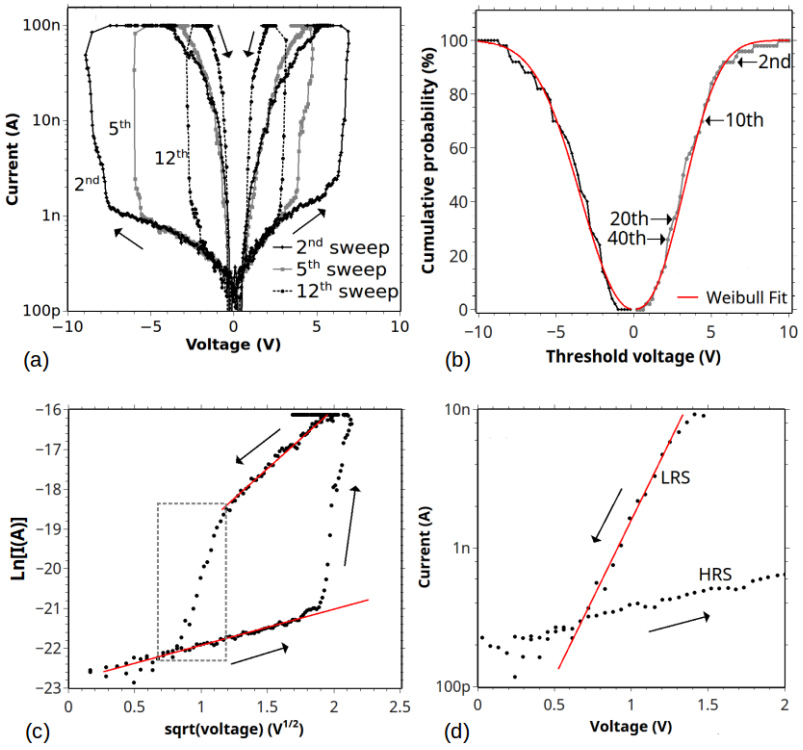


Figure 6.2: (a) I-V plot taken from the device with 100 nA current limit imposed during measurement and showing hysteresis with (b) cumulative probability plot of positive/negative threshold voltages fitted using Weibull's function [22]. The positive threshold voltages are shown for the 2nd, 10th, 20th and 40th sweeps. (c) and (d) show fitting of HRS and LRS in the 4th I-V sweep (see main text).

Figure 6.3(a) shows a back-scattered electron image of the device after the forming sweep shown in Figure 6.1(d) and the subsequent sweeps producing the systematic resistive switching shown in Figure 6.2. A fragmented filament extends across the a-ZTO layer between the planar Pt capped Ag electrodes (brightest areas left and right within the image). The contrast between the fragments and surrounding a-ZTO in the image shows that they contain material with higher atomic number. Since Pt is considered inert and is not in direct contact with the a-ZTO, it is asserted that the filament is rich in Ag. Similar Ag filaments have been observed in

memristors based on a variety of oxides using high resolution transmission electron microscopy [14,15] but observation in a scanning electron microscope is notable because using solid substrates (not electron-transparent membranes or cross-sectioned specimens) is possible and greatly simplifies fabrication/handling. The filament must reside largely within the a-ZTO layer (sub-surface) since scanning probe microscopy (not shown) revealed no measurable alteration of surface topography in the area between the contacts. RMS roughness was 0.4 nm before and after operation.

Identical Ag/Pt contacts on an amorphous C film with similarly high sheet resistance failed to form a filament with four times the applied electric field, showing that the a-ZTO host layer plays an important role in the device characteristics. As shown in Figure 6.1(c), a similar device with Pt contacts (and no Ag layer) patterned on the same a-ZTO film exhibited less significant I-V hysteresis under similar measurement conditions. The current in that device also failed to reach the 100 nA limit imposed. Based on the measurements, it is proposed that a local current path exists between the contacts in the pristine semi-insulating a-ZTO. In a study on Ag migration in TiO_2 using density functional theory (DFT) [24], Prada et al. state that mobile defects such as oxygen vacancies are likely to facilitate initial conduction. In a-ZTO transistors with AgO_x gates, improved behaviour after bias stress has been attributed to the motion of these defects at higher fields [4]. It is suggested that as the field is increased in the memristor and the current through the local path increases during the forming process, Ag ions migrate along the conduction path with assistance given by Joule heating, vacancy motion and/or bond rearrangement. This work and preceding work [4] suggests that in a-ZTO, as in TiO_2 [24], Ag ions are stable. Application of electric field is therefore an effective means to arrange Ag ions/clusters within a-ZTO to form filaments. The discrete Fourier transform (DFT) calculations

of Prada et al. [24] showed steric repulsion of Ag within TiO_2, leading the authors to state that amorphisation of the host material may assist in further mobilising Ag ions [24]. Hence, increased ion mobility is expected in a-ZTO and likely to cause faster switching and faster filament relaxation.

6.5. Microscopy observation of memristive silver filaments

Devices with identical layer structure to that shown in Figure 6.1(b) but with differing electrode separations were fabricated, operated and imaged to further investigate the behaviour of the conducting Ag filaments in a-ZTO. Due to the presence of the Schottky barriers (and associated potential drops) at the electrode/a-ZTO interfaces, the threshold voltage did not increase linearly with the electrode separation. The threshold voltage range for filaments with lengths 200 - 700 nm was 24 – 38 V. The electrical characteristics exhibited by the a-ZTO memristors are consistent with those of nanoscale conducting wires subject to Rayleigh instability [25]. Figure 6.3(b) shows a device with electrodes separated by 200 nm after operation with the same 100 nA current limit and sweep parameters as employed for the device in Figure 6.3(a). (The I-V characteristics of this device are shown in Figure 6.4). The filament shown in Figure 6.3(a) has clearly ruptured and has done so in a manner consistent with the Rayleigh instability condition. The rupture in the shorter filament in Figure 6.3(b) is not resolvable in the SEM image but since this filament also exhibited volatile behaviour and was in the HRS before imaging, it is believed to be discontinuous. If the characteristic Rayleigh perturbation wavelength exceeds the circumference of a filament (assumed cylindrical), then it can rapidly fragment (as in Figure 6.3(a)) [25]. If the opposite is true and the circumference of the filament exceeds the Rayleigh perturbation wavelength, then the filament is more robust (as in Figure 6.3(b)) [25]. Longer, thinner filaments are therefore likely to relax (fragment) more quickly and exhibit short

retention times. Additional images of device filaments are shown Figure 6.5. Images processed to extract parameters related to the Rayleigh instability conditions of the filaments are provided in Figure 6.6.

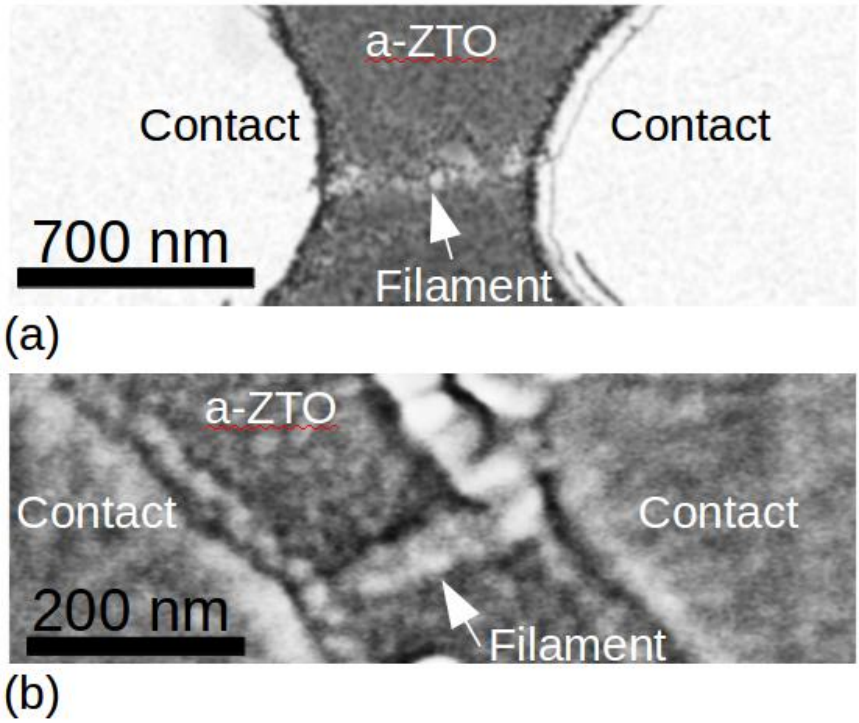


Figure 6.3: Back-scattered electron image of nanoscale filaments extending between Pt-capped Ag electrodes separated by (a) 700 nm and (b) 200 nm.

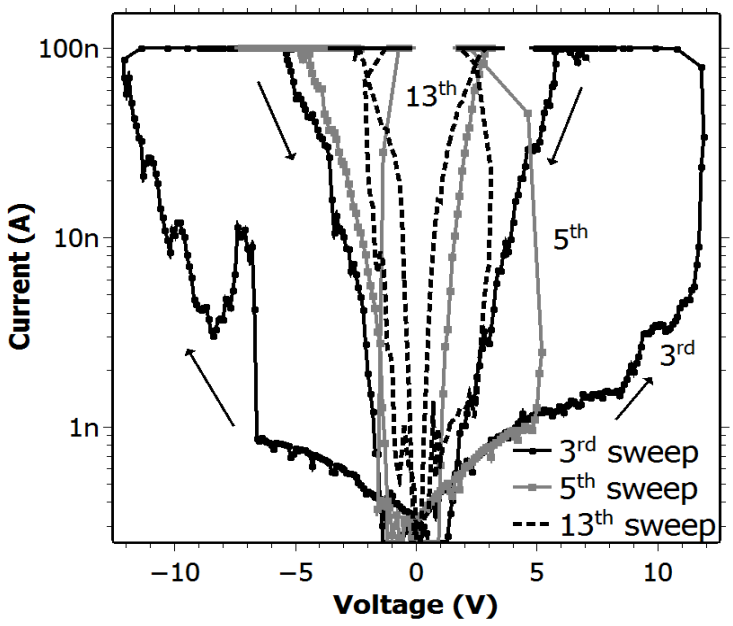


Figure 6.4: I-V characteristic of the lateral memristor with 200 nm electrode separation. Reductions in threshold voltage and increased conductance in LRS occur with increased activity (sweep number). This behaviour was common to all devices tested and with electrode separations up to 700 nm.

Nanoparticles formed due to Rayleigh instability obey the following relationship (as determined by Nichols & Mullins [27]),

$$\lambda = 8.89L \tag{6.2}$$

and

$$0.5d = 3.78L \tag{6.3}$$

where L is the radius of the filament prior to nanoparticle formation. Therefore, if nanoparticles are formed due to Rayleigh instability, $\lambda / 0.5d = 4.7$. Plots of $\lambda / 0.5d$ in the inset of Figure 6.6 (a & b) show that the Rayleigh criterion is satisfied irrespective of electrode spacing.

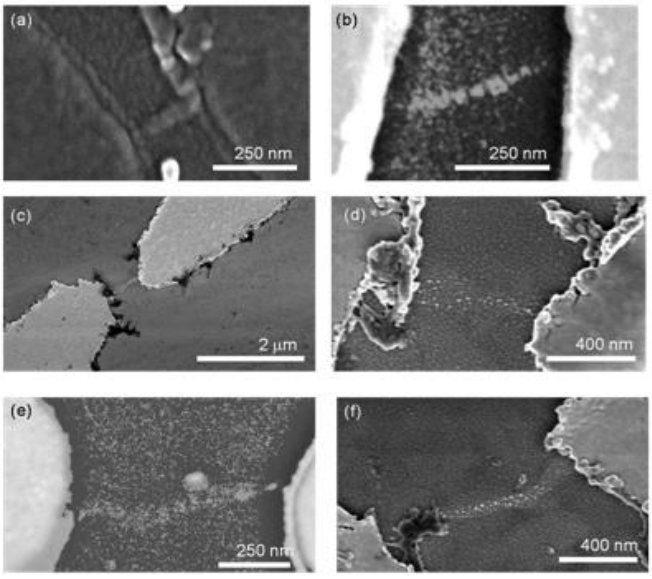


Figure 6.5: Scanning electron microscope (SEM) images of embedded Ag nanoparticle filaments in laterally configured Ag/ZTO/Ag memristors with electrode separations of (a) 200 nm, (b-d) 300 nm and (e,f) 700 nm.

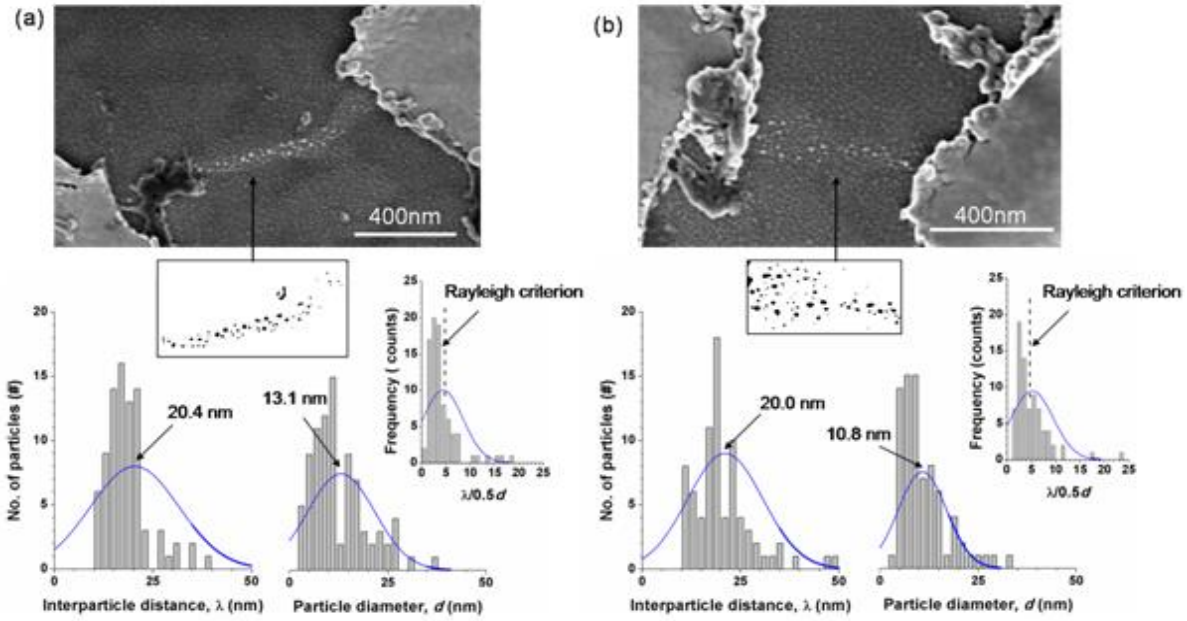


Figure 6.6: Interparticle distance (λ) and particle diameter (*d*) distributions of embedded Ag nanoparticle filaments in laterally configured Ag/ZTO/Ag memristors with electrode separations

of (a) 700 nm and (b) 300 nm. The ratio between λ and the particle radii (0.5 *d*) are also included. Nanoparticle identification was achieved using the software ImageJ [26]. SEM images were inverted and then manually adjusted using a 1-bit threshold filter. The interparticle distance was analysed using the ImageJ plugin by Y. Mao (Next Nearest Neighbour Distances) [26].

As more current is passed through the Ag-rich filaments in the devices, they become more conductive (as shown later in Figure 6.8). This is attributed to increased migration of Ag (from the contacts into the filament) brought about by repeated activity and/or by increased device current [13]. The Rayleigh instability model also predicts that for a filament of fixed length, prolonged relaxation accompanies increased radius [28]. The range of relaxation times (defined as time taken to fall to 50% of conductance in LRS) was 100 - 300 ms for filaments with lengths ranging from 200 - 700 nm. Measurements of relaxation times are plotted in Figure 6.7 and are comparable with many threshold switching devices incorporating Ag or Cu filaments [13]. Other authors have shown that metallic filaments undergoing Rayleigh instability exhibit a linear relationship between the fourth power of their radius and their relaxation time [29]. The data seems to follow this relationship (see Figure 6.7). This is surprising given the barrier limited transport shown in Figure 6.2(c) which would suggest that the filaments are far from metallic.

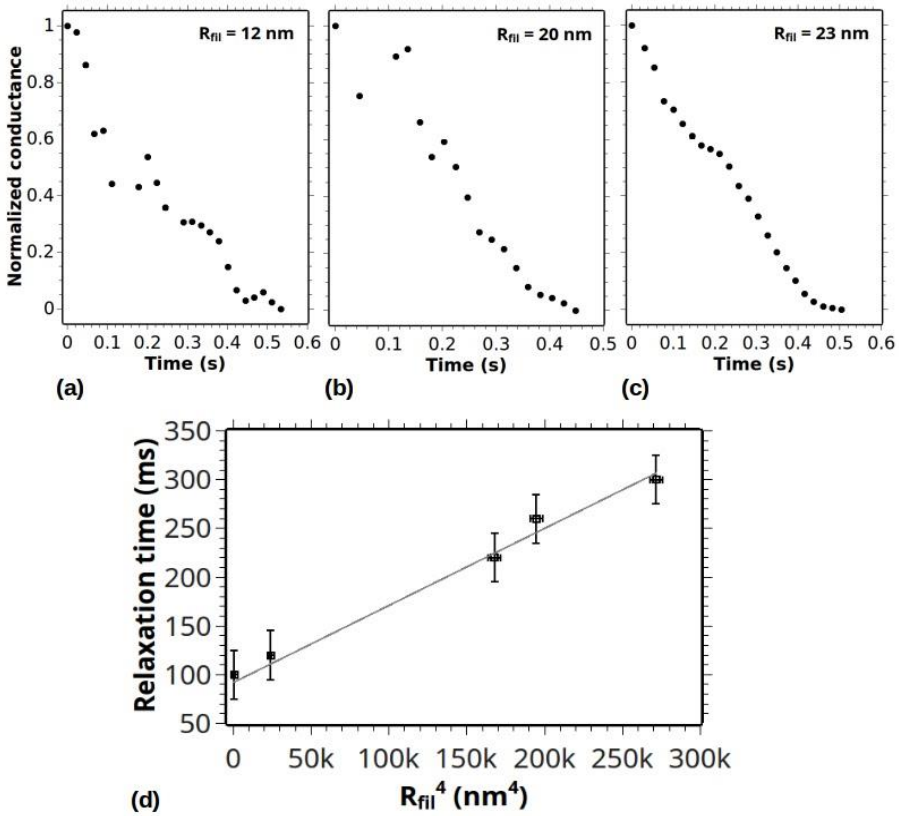


Figure 6.7: Typical relaxation characteristics following LRS of memristors with filaments of radius (R_{fil}) (a) 12, (b) 20 and (c) 23 nm, (d) shows that this data follows the relationship often observed in nanoscale filaments subject to Rayleigh instability (see main text). Error bars incorporate variance between repeated measurements and measurement uncertainty.

6.6. Emulating synaptic behaviour

Figure 6.8 shows the evolution of the conductance of a device with a 700 nm filament with repeated I-V sweeps. Current limits of 100 nA (Figure 6.8(a)) and 1 μA (Figure 6.8(b)) were imposed and the conductance was measured at half the current limit in the LRS. Both plots are characterised by an initial increase in conductance then saturation. This behaviour, routinely observed in biological synapses, is critical for memory and learning. Its emulation by inorganic

memristive devices has been reported by numerous authors (see for example [30]). With the lower current limit imposed, the conductance is highly stochastic (Figure 6.8(a)) consistent with a filament that is unstable on a timescale equivalent to one sweep. Nevertheless, saturation is evident and reasonable agreement is achieved between the experimental data and the grey curve in the plot. This curve follows $G = G_0 + G_{sat}(1 - Ae^{-n/B})$ where G is the device conductance, G_0 is the conductance in the unformed (pristine) state, G_{sat} is the saturation conductance, n is the sweep number and constants A = 1.5 and B = 5. After the forming process, G_0 represents the comparatively small parallel conductance between the electrodes through the unmodified a-ZTO. Similar phenomenological equations have been used to describe activity dependent biological synaptic efficacy [31] and conductivity in inorganic memristors [32]. Allowing a greater current through the device increases the saturation current and reduces the scatter in the conductance data (Figure 6.8(b)). Once again, the saturation behaviour follows the model previously used to describe synaptic behaviour with A = 1.5 and B = 2. The inset in Figure 6.8(b) shows the LRS I-V characteristic during the 6th sweep. In contrast to the LRS I-V characteristics with the lower current limit imposed, a single current limiting mechanism describes this I-V relationship. The square law dependence shown in the inset is consistent with space charge limited conduction resulting from carrier injection into an insulating material. The emergence of this characteristic implies that at higher compliance the junctions/interface layers at the electrodes are no longer the strongest influence on transport. When the higher current limit is imposed, the higher saturation conductance observed is the result of a more substantial filament (with larger radius). It is suggested that the Ag/a-ZTO junctions are effectively ‘shorted out’ by this more substantial filament.

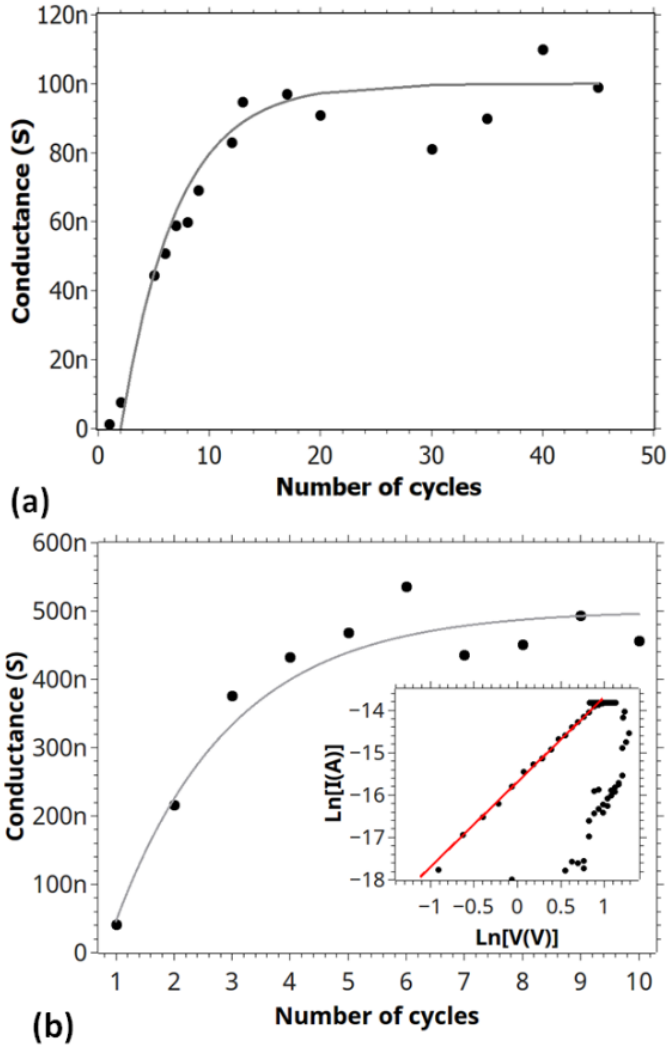


Figure 6.8: Conductance of the filament (determined at half the current limit) in the LRS versus number of cycles with current limits of (a) 100 nA and (b) 1.00 µA imposed. The grey curves follow a model equation for saturation in biological synapses (see main text). Inset in (b) shows that after the sixth sweep, transport in the LRS is consistent with space charge limited conduction.

The structure of the memristor reported here was chosen to facilitate microscopy following operation without additional sample preparation. Achieving the highest on:off resistance ratio, longest retention times and greatest stability were not aims of the study. Whilst it

has been claimed that solid state synapses should exhibit linear potentiation and depression to faithfully mimic biological synapses, numerous authors have demonstrated memristors that relax spontaneously and perform useful bio-inspired signal processing or pattern recognition tasks, see for example [33,34]. The resistive switching endurance of these devices (shown in Figure 6.9) is however promising and further experiments with encapsulated devices are planned. All devices tested exhibited synaptic plasticity and conduction saturation, both of which are widely acknowledged to be crucial in biological memory. Aside from this aspect of their behaviour, their fast relaxation and the significant LRS:HRS ratio show that there is potential for devices like these to serve as threshold switching selectors. These enable selective programming of individual memristors within arrays [13].

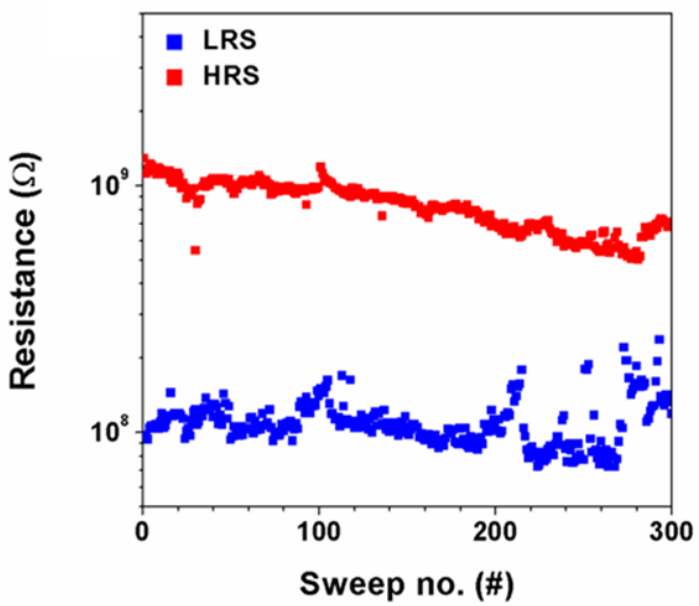


Figure 6.9: Resistive switching endurance of a volatile, filamentary Ag/a-ZTO lateral memristor. These characteristics were measured in air at room temperature.

6.7. Conclusion

Planar Ag electrodes supported on amorphous zinc-tin-oxide (a-ZTO) and separated by differing sub-micron separations have exhibited volatile resistive switching and activity dependent conductance after the formation of Ag-rich nanoscale filaments. Whilst Ag filaments have been reported to form in a long list of materials, it has not been found a-ZTO included in this list, despite the effects of Ag migration being observed in Ag gated a-ZTO transistors. Current-voltage measurements performed on the Ag/a-ZTO memristors showed that in the pristine state, the Pt/Ag contacts formed back-to-back Schottky contacts to the underlying a-ZTO. In the HRS state, current transport was consistent with Schottky emission. In the LRS, the filament exhibited transport behaviour and threshold voltage that depended both on activity and on the current limit imposed. The filament was subject to Rayleigh instability and the LRS relaxation time was shown to increase when more current was passed through the device and the effective radius of the filament increased. Saturation of the device conductance after repeated I-V sweeps resembled biological synaptic behaviour. This work highlights the potential for transparent neuromorphic circuits combining a-ZTO-based TFTs and a-ZTO-based memristors.

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Chapter 7

Discussion, conclusion and future work

This chapter first provides a summary of the works presented in chapter 4, 5 and 6. It then follows with a discussion of important findings in the context of previously published work, including graphitic carbon as electrical contacts to semiconducting devices, a-ZTO thin film synthesis and ZTO resistive switching devices. Finally, conclusions of the thesis are presented, and further work is proposed.

7.1. Energetic deposition, measurement and simulation of graphitic contacts to semiconductors

In chapter 4, the microstructural and electrical properties of graphitic carbon contacts to 6H-SiC were presented. Carbon films were energetically deposited using a FCVA system at room temperature and elevated temperature (300 ^{o}C) with low and high biases (from 100 to 700 V). XPS was used to determine sp^2 fractions of the C films. It was found that as the substrate bias increased, the sp^2 fraction increased and rectifying device characteristics emerged. At 700 V, the rectification ratio at ± 2V exceeded four orders of magnitude. An ideality factor of 1.7 was determined from the measurements. Lower deposition energy resulted in weakly rectifying junctions with larger series resistance. Cross-sectional TEM imaging revealed the absence of an oxide layer between the C anodes and the SiC substrate. Since no oxide removal process (such as etching in Hydrofluoric (HF) acid) was applied prior to the deposition, the lack of native oxide at the interface was attributed to sputtering by energetic C ions during the deposition. TEM imaging and EELS (Electron energy loss spectroscopy) showed the presence of an interfacial layer ~ 3 nm in thickness between the

graphite anode and the 6H-SiC substrate. This interlayer was an amorphous and carbon-rich layer. This interface layer was beneficial to the device characteristics. Simulations suggest a highly resistive thin interlayer significantly reduces reverse leakage current. The forward current is largely unaffected by the presence of the interlayer.

Finite element modelling using Sentaurus TCAD was adopted to further study the characteristics of junctions between carbon contacts and 6H-SiC. Since barrier height inhomogeneity is a well-known problem affecting device performance, it was enabled in TCAD model by dividing the diode into two equal smaller areas (two diodes in parallel with different barrier heights). Several important parameters, such as electrode work function, interfacial layer thickness, oxide thickness and series resistance, were included in the simulations to provide the best fit with the electrical measurements and electron microscopy. Since carbon materials were not available in the standard TCAD library, new material files were written using parameters form the literature. The models that provided best agreement between simulated and experimental I-V characteristics are presented in Table 4.3. Variations in electrode work functions contribute to extracted ideality factors that exceed unity. Simulation results also confirmed that the presence of the interfacial layer between the anode and the substrate is important to device characteristics. When comparing the I-V characteristics from the models with and without the presence of a 3 nm thick a-C interface layer, the device with no a-C interface layer showed a reduction in the rectification ratio by three orders of magnitude. The device simulations also suggest that improved rectification ratios may be obtained from C/6H-SiC Schottky junctions if their lateral inhomogeneity is improved and/or the thickness/properties of interfacial a-C/SiC mixing layers are more accurately controlled.

Further work has been done to investigate the properties of the interfaces between graphitic contacts and semiconductors by performing temperature dependent electrical (I-V-T) measurements [1]. In this work, graphitic contacts were energetically deposited onto p-Si substrates with bias/temperature of 1.0kV/100 ^{o}C. From room temperature I-V measurements, the extracted barrier height and ideality factors were 0.6 eV and 1.05, respectively. The current rectification ratio (at ±1 V) exceeded five orders of magnitude and these devices are amongst the best reported for graphitic/graphemic junctions to silicon. For comparison, 'ideal' graphene-Si Schottky diodes have been reported with ideality factor of 1.08 [2]. From the I-V-T measurement, the effective Richardson constant can be obtained from the ($ln\left(\frac{I_S}{AT^2}\right) = -\frac{q\Phi_{Beff}}{kT} + \ln(A^*_{eff})$) plot [3]. Extrapolating to the intercept gives an effective Richardson value of 27 ± 8 $Acm^{-2}K^{-2}$, which is in great agreement with the theoretical value of 32 $Acm^{-2}K^{-2}$ for holes in p-type Si [4]. Since no alteration was observed in the value of the extracted Richardson constants, it indicates that energetic deposition process does not introduce significant damage to the Si substrate. Hence, this energetic deposition method can provide high-quality contacts necessary for Schottky diodes, by sputtering away surface contamination/native oxide layers without incurring damage to the underlying substrate.

7.2. Characterisation and doping of a-ZTO using HiPIMS

In chapter 5, the characteristics of a-ZTO deposited energetically using HiPIMS were reported. This sputtering technique has proved capable of producing high quality metal oxide layers suitable for electronic devices [5, 6]. Reactive co-deposition from Zn (HiPIMS mode) and Sn (DC magnetron sputtering mode) targets yielded a-ZTO with varying Zn:Sn composition across a 4-inch diameter sapphire substrate. The compositional range achieved

in the film was Zn:Sn = 27:75 – 54:46. The structural, electrical and optical properties of this film were then studied as a function of composition.

As-deposited, the films were amorphous, transparent and semi-insulating. Diffraction patterns collected from a TEM cross-sectioned sample showed an amorphous structure, in agreement with XRD collected from the same sample to be 70 nm. The film thickness was measured from step-height measurements using AFM. The film was smooth with the RMS roughness being less than 1 % of the film thickness. The films were all transparent with transmission over 80 % in the visible range. Tauc plots of the as-deposited a-ZTO films revealed a bandgap of ~ 2.9 eV. XPS measurements were performed to further study the surface chemistry of the as-deposited a-ZTO films. It was found that in the O 1s spectra of the ZTO films, together with the lattice oxygen peak, there was an additional surface-bound OH peak. This is problematic when forming Schottky contacts on the a-ZTO films since OH coverage causes electron accumulation, leading to a downward band-bending at the surface. Removal of this surface contamination was possible by surface etching inside the XPS system (using Ar ions with average energy of 200 eV). It has been shown that during energetic deposition of carbon contacts on p-Si and 6H-SiC, the surface contamination (native oxide layer) can be removed by sputtering without causing any significant damage to the substrate itself. Hence, there is potential for graphitic carbon to be deposited energetically onto the a-ZTO films, for OH removal and improved Schottky barrier devices.

Hall measurements showed the as-deposited films were highly resistive with the resistivity exceeding 1 kΩ.cm. As outlined in Section 5.4.4, hydrogen bound within oxygen vacancy sites has been reported to be a stable and active defect leading to n-type doping in ZnO [7, 8]. Annealing in the presence of hydrogen has been shown to increase carrier concentration in ZnO and SnO_2. In this study, as-deposited a-ZTO films were annealed in

hydrogen atmosphere (100 mTorr) at 500 ^{o}C for 1 hour to increase their conductivity. The annealed samples were then re-characterised and compared with the as-deposited samples. After annealing, all samples remained amorphous but exhibited a significant decrease in resistivity, moderate n-type carrier concentrations (~ 10^{17} cm^{-3}) and mobility up to 13 cm^2/V.s. The results indicate that annealing in hydrogen leads to improved and stable electronic properties with no adverse effects on microstructure of the ZTO films. The results demonstrate that HiPIMS can produce dense and high quality amorphous ZTO, including low roughness, low optical absorption and with controlled doping possible by hydrogen annealing. The material is therefore suitable for applications including transparent thin film transistors.

7.3. Device application for HiPIMS a-ZTO: memristive silver filaments

Chapter 6 reports the observation and characterisation of memristive silver filaments in a-ZTO. The application of ZTO in devices has been largely continued to thin-film transistors. Combining memristors and transistors for neuromorphic circuits has been gaining interest and demonstrated using Si-based devices [9]. If it was investigated using AOS materials, more benefits would be added to these devices/circuits since transparency and/or flexibility would be gained.

It has been shown that partially oxidised Ag can form highly rectifying contacts to a-ZTO and serve as a gate material in metal-semiconductor field-effect-transistors (MESTETs). While high stability is reported under normal operating voltages/fields [10], at higher fields Ag has been observed to migrate into and through the a-ZTO, suggesting potential for memristive behaviour in this material combination.

In chapter 6, lateral memristors consisting of planar Ag electrodes (with sub-micrometre separation) supported on thin films of a-ZTO (70 nm) were characterised. Lateral devices were constructed to enable observation of the region between the contacts before and after electrical characterisation using electron microscopy, and to thereby better understand the underlying switching mechanisms.

The devices were written using electron beam lithography with gaps between two electrodes varied from 200 to 700 nm. 30 nm of Ag was then deposited to serve as electrodes and followed by another 30 nm of Pt as a capping layer. As deposited, the resistivity of the a-ZTO exceeded 10 kΩ.cm. At low bias (below the threshold voltage), the devices remained in a high resistance state and the I-V relationship was linear. This agrees with the characteristics of back-to-back (AgO_x/a-ZTO/AgO_x) Schottky diodes since the main transport mechanism is due to tunneling of carriers through the barriers. As the electric field increased, the current increased and the filament formed at 0.5 MV/cm, corresponding to a voltage of ~ 38 V with a 700 nm – gap device. Above threshold voltages and with a 100 nA current limit imposed, switching between high and low resistance states was observed. The devices were volatile in their state retention behaviour. As the number of sweeps increased, threshold voltages decreased, suggesting that the device conductance depended on prior activity. Fitting of HRS and LRS of a typical sweep was also performed to study the transport mechanisms of the device. In the HRS, I-V characteristics indicated transport by Schottky emission and this model enabled accurate prediction of the relative permittivity of a-ZTO. In the LRS, the transport was still governed by Schottky emission. Using this model and assuming that the permittivity of the a-ZTO films remained constant, a new effective gap between the contacts was found to be $1/10^{th}$ of the original gap (70 nm). Ohmic conduction was not observed in the LRS, likely due to an incomplete filament between the electrodes caused by gaps and/or inter-granular barriers.

Electron micrographs of the formed lateral memristors revealed nanoscale filaments extending between two electrodes. These filaments were fragmented and consistent with the volatile behaviour of the devices and attributed to the Rayleigh instability. The contrast between the filaments and surrounding a-ZTO in the image showed that they contain an element with higher atomic number than Zn, Sn or O. This suggested the filaments were not caused by migration of oxygen ions. In addition, identical devices fabricated on the same a-ZTO films but with Pt contacts did not switch. It is therefore believed that the filament is mainly formed from Ag due to field induced migration of Ag ions.

Devices with differing electrode separations were fabricated, operated and imaged to further investigate the behaviour of the conducting Ag filaments in a-ZTO. The threshold voltage decreased as the gap between the two contacts decreased. Rupture in filaments was observed in all devices, consistent with the Rayleigh instability condition [11]. According to this model, longer and thinner filaments fragment more quickly and exhibit shorter retention times. This was observed experimentally. For a given electrode separation, longer relaxation times correspond to larger effective radius of the filament [12]. The range of relaxation times observed from the Ag/a-ZTO devices was 100-300 ms for filaments (and contact separations) ranging from 200 to 700 nm.

As more current passed through the devices, they became more conductive. The conductivity increased with increasing sweep number but quickly saturated. Saturation of the device conductance followed a phenomenological function used to model activity-dependent biologic synaptic behaviour [13]. This work highlights the potential for transparent neuromorphic circuits combining a-ZTO-based and a-ZTO-based memristors.

7.4. Conclusion

The work described thus far has shown that energetic deposition is capable of producing materials suitable for electronic device applications. The additional energy in the depositing flux has been found beneficial in the deposition of a-ZTO. The results presented here support the assertion that deposition at elevated energies promotes adhesion and assists in preventing crystallisation in the growing films [14]. The bonding fractions of carbon films are sensitive to the energy of deposition and by varying from lower energy to energies of around 1 keV, thin films can be produced with electrical properties varying from highly insulating to semi-metallic [15]. In this work, films with high sp^2 fractions have been deposited to serve as stable electrical contacts. Here too, benefits of the elevated deposition energy have emerged. The energetic plasma is beneficial in removing surface contaminants to produce a high-quality interface with improved lateral homogeneity. This has enabled high quality Schottky contacts to be formed to 6H-SiC (and other semiconducting materials). The work reported in this thesis has therefore shown that energetic deposition offers advantages in forming semiconducting oxide layers and in forming device contacts. The technique is well established in the tribological coatings industry but potential exists for it to be exploited in producing electronic materials including those required in next-generation technologies such as flexible electronics and neuromorphic electronics.

7.5. Future work

7.5.1. Carrier transport mechanisms in a-ZTO

Resistivity-temperature (R-T) measurements have been used to study transport mechanisms in a-ZTO films. R-T measurements were performed using a temperature controlled Linkam probe-stage. Temperatures were varied from 90 K to 450 K, while the

resistance of the active channel between source and drain contacts was measured, as shown in Figure 7.1.

Figure 7.2 shows that the resistivity does not follow a simple thermally activated behaviour since the Arrhenius plots exhibit large deviations from straight lines, especially at high temperature (> 160 K). At low temperatures, a better straight line was obtained in the log R vs $T^{-1/4}$ plot in the range of 90 K to 130 K. It is believed in the low temperature regime, the hopping transport mechanism, or variable-range hopping (VRH) takes place by tunneling (hopping) of charge carriers between impurity atoms [16]. This transport mechanism is a well – known conduction model caused by localized states in disordered or highly doped semiconductors [17]. However, further investigation is required to understand the complex modes of transport, especially in the higher temperature regime.

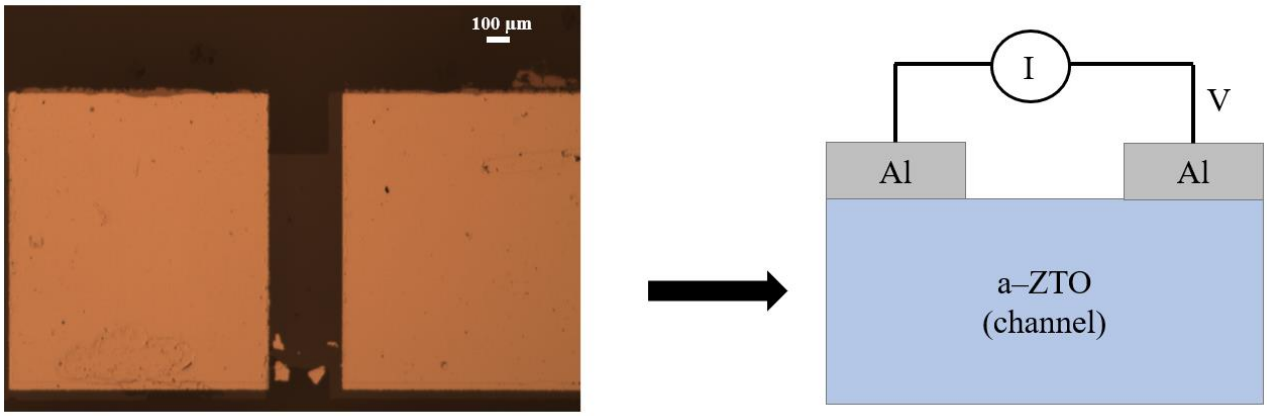


Figure 7.1: Optical image of the device showing source and drain contacts, and schematic of two probe R – T measurement

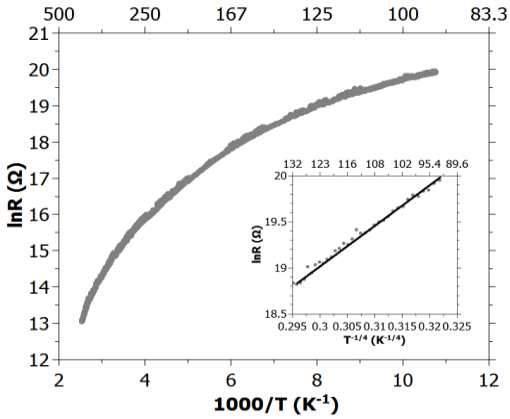


Figure 7.2: Arrhenius plot of resistivity for a-ZTO films. Inset shows a linear fit of log R vs $T^{-1/4}$ at low temperature, suggesting the VRH transport mechanism

7.5.2. Metal-insulator-semiconductor field effect transistors (MISFETs) on HiPIMS a-ZTO films

Bottom gate TFTs based on a-ZTO were fabricated. Highly doped Si and thermally grown SiO_2 (50 nm) served as gate electrode and insulator, respectively. The device structure was defined by photolithography and lift-off. As a channel layer, 70 nm thick ZTO films were co-deposited using the same conditions as previously reported in chapter 5. Annealing in hydrogen atmosphere (1 hr, 500 ^oC, 100 mTorr H_2) was conducted to introduce doping to the channel. Hall measurement using an Ecopia HMS-3000 system revealed the bulk carrier concentration of the annealed 1 cm × 1 cm films to be 5×10^{18} cm^{-3}. Aluminum 300 nm source and drain contacts were then thermally evaporated on top of the ZTO. The channel width (W) and length (L) were 300 and 100 µm, respectively. Device characteristics were measured using a Keysight B2902A at room temperature in dark conditions.

Figure 7.3 shows a cross-sectional illustration of the ZTO MISFET structure employed here. The output characteristics in Figure 7.4 show the value of drain current I_{DS} increased as the drain bias increased from 0 to 15 V. Positive gate bias V_{GS} increased I_{DS} to

0.35 mA as V_{DS} increased to 15 V, whereas negative V_{GS} decreased I_{DS} and depleted carriers in the channel. The field-effect mobilities were extracted at V_{DS} = 1 V to ensure that they are from the linear region and calculated from $\mu_{lin} = \frac{\partial I_D}{\partial V_{GS}} \times \frac{L}{W C_0 V_{DS}}$ (see Figure 7.5) [18,19], where C_0 is the gate capacitance per unit area. The mobility for the ZTO devices presented here was ~ 0.6 cm^2/V.s and smaller than the Hall mobility (15 cm^2/V.s).

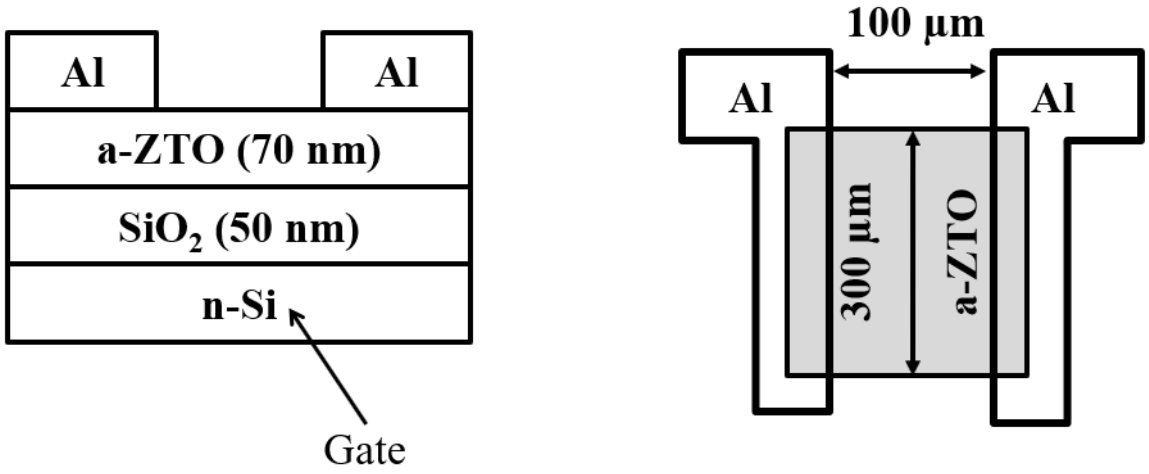


Figure 7.3: Cross sectional and plan view of a bottom-gate a-ZTO TFT

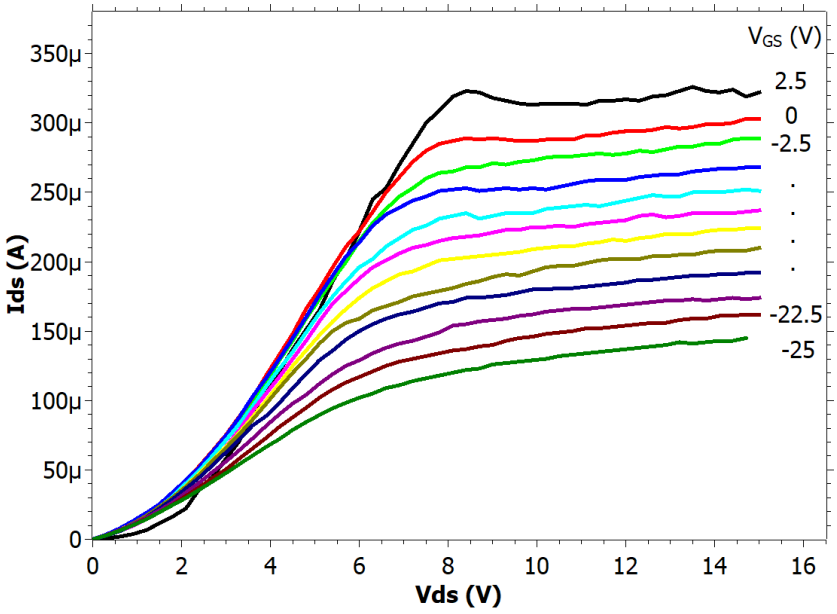


Figure 7.4: Drain current-drain voltage (I_{DS} - V_{DS}) output characteristics for a ZTO MISFET structure (described in Figure 7.3)

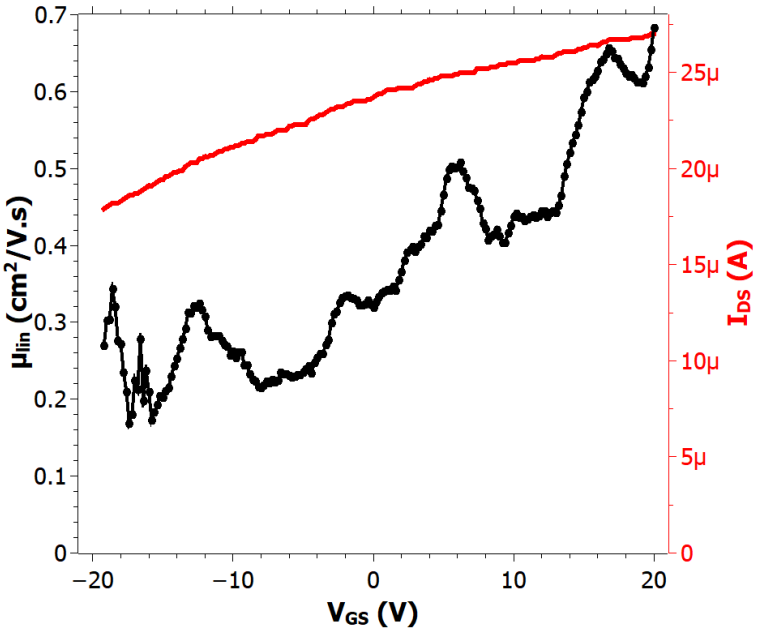


Figure 7.5: Transfer characteristics and field-effect mobility in the linear region (μ_{lin}) as a function of V_{GS} at V_{DS} = 1 V a ZTO MISFET structure (described in Figure 7.3)

However, since the a-ZTO films used for this initial investigation were thick (70 nm) and doped to ~ 10^{17} cm^{-3}, the transistors could not be turned off. Further optimisation of the device fabrication process is expected to improve device performance. Suggested future work is therefore to produce thinner a-ZTO transistor channel layers to ensure full channel depletion. TCAD simulations have assisted in determining that an estimated film thickness of 50 nm is necessary to ensure the transistor can be switched off at lower fields (< 2 MV/cm).

7.5.3. Flexible a-ZTO TFTs with graphitic contacts

Graphitic carbon contacts have received much attention recently. Carbon has been demonstrated to form high-quality and stable rectifying contacts to a wide range of semiconductors, including Si [20], SiC [20], GaAs [20], GaN [21] and ZnO [22,23]. These contacts have either been painted from colloidal graphite [20,22] or thermally evaporated [23]. These low energy deposition techniques provide good contact to the semiconductor surface. However, as stated, in ZnO-based materials including ZTO, hydroxyl groups attached on the ZnO surfaces are known to cause electron accumulation and downward band-bending which limit the performance of Schottky diodes [24]. These methods require surface treatment such as using oxygen plasma [25] prior to the deposition of rectifying contacts, to remove OH attachment and improved Schottky barrier devices.

Recent works have shown that energetic deposition of graphitic contacts to p-Si [26] and 6H-SiC [27,28] remove surface contaminants without introducing significant damage to the underlying substrates. It is, therefore, believed that energetically deposited graphitic carbon onto the a-ZTO films can help to remove OH groups present on the a-ZTO surfaces and to improve the device performance. In future work, energetically deposited carbon will be employed as both a gate insulator (t-aC) and as gate electrodes (graphitic C) to investigate the potential of carbon as a multifunctional material in a-ZTO TFTs. Thus, combining carbon

and a-ZTO on a flexible substrate should yield high performance, flexible TFTs. The device structure proposed is shown below in Figure 7.6.

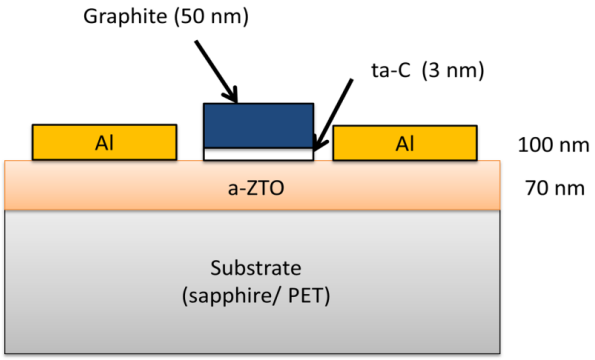


Figure 7.6: Device structure proposed for future a-ZTO MISFETs

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